

Week 1: From Research Question to Statistical Model

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Skills Learning – Lecture

This week, we are shifting from basic R skills toward statistical thinking.

In the introductory course, we focused on importing data, exploring data, cleaning columns, summarizing variables, and making plots. In this advanced course, we will build on those skills and ask a new question:

Given my biological question and my data, what kind of statistical model should I use?

Before we can choose a model, we need to understand:

- what kind of outcome variable we have
- whether observations are independent or repeated
- whether the dataset is cross-sectional or longitudinal
- what the distribution of the response variable looks like
- whether the predictor variables are continuous, categorical, count, binary, or proportions

This week is about building that foundation.

0. Load Required Packages

```
# install packages only if you have not yet  
# -----  
# install.packages("tidyverse")  
# install.packages("janitor")  
# install.packages("here")  
  
# Load packages  
library(tidyverse)  
library(janitor)  
library(here)
```

1. Load and Preview the Dataset

Today we are using a faux gorilla hormone + behavioral dataset.

Each row represents one sample/observation from a gorilla. Some gorillas appear more than once, which means this dataset contains **repeated measures**.

The dataset includes:

- gorilla identity and group information
- sex and age class
- hormone values
- behavioral variables
- health-related variables
- ecological variables

```
# check that you are in the correct working directory
getwd()
```

```
## [1] "/Users/rsteinitz/Documents/github/R Advanced Data Analysis/Week 1"
```

```
# list files and folders in the current working directory
list.files()
```

```
## [1] "gorilla_hormone_behavior_data.csv" "Radac_wk1.pdf"
## [3] "Radac_wk1.Rmd"
```

```
# list files inside the Week 1 folder
list.files("Week 1")
```

```
## character(0)
```

```
# Load dataset
data <- readr::read_csv(here("Week 1", "gorilla_hormone_behavior_data.csv")) %>%
  janitor::clean_names()
```

```
## Rows: 240 Columns: 19
## -- Column specification -----
## Delimiter: ","
## chr   (7): sample_id, gorilla_id, gorilla_name, group, observer, sex, age_class
## dbl   (11): age_years, dominance_rank, group_size, fruit_availability_index, ...
## date   (1): sample_date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
# Preview the structure
glimpse(data, width = 80)
```

```
## Rows: 240
## Columns: 19
## $ sample_id      <chr> "G001_S01", "G001_S02", "G001_S03", "G001_S04~
## $ gorilla_id     <chr> "G001", "G001", "G001", "G001", "G001", "G001~
## $ gorilla_name    <chr> "Gasore", "Gasore", "Gasore", "Gasore", "Gaso~
## $ group           <chr> "BWE", "BWE", "BWE", "BWE", "BWE", "BWE", "BW~
## $ sample_date     <date> 2025-05-23, 2025-08-18, 2025-09-10, 2025-02--
## $ observer        <chr> "Mark", "Sarah", "Mary", "Sarah", "Mark", "Ro~
## $ sex             <chr> "F", "F", "F", "F", "F", "F", "F", "F", "F", ~
## $ age_class        <chr> "adult", "adult", "adult", "adult", "adult", ~
## $ age_years        <dbl> 14.2, 14.2, 14.2, 14.2, 14.2, 14.2, 14.2, 14.~
```

```
## $ dominance_rank      <dbl> 8, 8, 8, 8, 8, 8, 8, 8, 6, 6, 6, 6, 6, 6, ~
## $ group_size          <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 3, 3, 3, 3, 3, 3, ~
## $ fruit_availability_index <dbl> 0.4040977, 0.3301009, 0.3134359, 0.7098761, 0~
## $ feeding_time_prop   <dbl> 0.2554745, 0.6272515, 0.5106832, 0.3779914, 0~
## $ aggression_events   <dbl> 2, 0, 1, 0, 0, 1, 1, 0, 3, 0, 1, 0, 0, 1, 0, ~
## $ parasite_load       <dbl> 28, 3, 12, 19, 15, 7, 38, 26, 6, 17, 10, 4, 5~
## $ respiratory_signs    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ injury_status        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
## $ cortisol_ng_ml       <dbl> 19.158612, 22.606257, 28.816896, 9.639142, 10~
## $ testosterone_ng_ml   <dbl> 5.717721, 3.666248, 3.084256, 4.783930, 3.218~
```

```
head(data)
```

```
## # A tibble: 6 x 19
##   sample_id gorilla_id gorilla_name group sample_date observer sex age_class
##   <chr>      <chr>      <chr>      <chr> <date>      <chr>    <chr> <chr>
## 1 G001_S01 G001      Gasore      BWE   2025-05-23 Mark     F    adult
## 2 G001_S02 G001      Gasore      BWE   2025-08-18 Sarah    F    adult
## 3 G001_S03 G001      Gasore      BWE   2025-09-10 Mary     F    adult
## 4 G001_S04 G001      Gasore      BWE   2025-02-02 Sarah    F    adult
## 5 G001_S05 G001      Gasore      BWE   2025-05-29 Mark     F    adult
## 6 G001_S06 G001      Gasore      BWE   2025-11-26 Robert   F    adult
## # i 11 more variables: age_years <dbl>, dominance_rank <dbl>, group_size <dbl>,
## #   fruit_availability_index <dbl>, feeding_time_prop <dbl>,
## #   aggression_events <dbl>, parasite_load <dbl>, respiratory_signs <dbl>,
## #   injury_status <dbl>, cortisol_ng_ml <dbl>, testosterone_ng_ml <dbl>
```

```
# Check column names
```

```
names(data)
```

```
## [1] "sample_id"      "gorilla_id"
## [3] "gorilla_name"   "group"
## [5] "sample_date"    "observer"
## [7] "sex"            "age_class"
## [9] "age_years"      "dominance_rank"
## [11] "group_size"     "fruit_availability_index"
## [13] "feeding_time_prop" "aggression_events"
## [15] "parasite_load"  "respiratory_signs"
## [17] "injury_status"  "cortisol_ng_ml"
## [19] "testosterone_ng_ml"
```

`janitor::clean_names()` standardizes column names.

It replaces spaces, capital letters, and symbols with clean lowercase names and underscores.

2. Inspect Dataset Structure

Before modeling anything, we need to understand the structure of the dataset.

```
# Basic overview
```

```
glimpse(data, width = 80)
```

```
## Rows: 240
## Columns: 19
## $ sample_id      <chr> "G001_S01", "G001_S02", "G001_S03", "G001_S04~
## $ gorilla_id     <chr> "G001", "G001", "G001", "G001", "G001", "G001~
## $ gorilla_name    <chr> "Gasore", "Gasore", "Gasore", "Gasore", "Gaso~
## $ group           <chr> "BWE", "BWE", "BWE", "BWE", "BWE", "BWE", "BW~
## $ sample_date     <date> 2025-05-23, 2025-08-18, 2025-09-10, 2025-02--
## $ observer        <chr> "Mark", "Sarah", "Mary", "Sarah", "Mark", "Ro~
## $ sex             <chr> "F", "F", "F", "F", "F", "F", "F", "F", "F", ~
## $ age_class        <chr> "adult", "adult", "adult", "adult", "adult", ~
## $ age_years        <dbl> 14.2, 14.2, 14.2, 14.2, 14.2, 14.2, 14.2, 14.~
## $ dominance_rank   <dbl> 8, 8, 8, 8, 8, 8, 8, 8, 6, 6, 6, 6, 6, 6, ~
## $ group_size       <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 3, 3, 3, 3, 3, 3, ~
## $ fruit_availability_index <dbl> 0.4040977, 0.3301009, 0.3134359, 0.7098761, 0~
## $ feeding_time_prop <dbl> 0.2554745, 0.6272515, 0.5106832, 0.3779914, 0~
## $ aggression_events <dbl> 2, 0, 1, 0, 0, 1, 1, 0, 3, 0, 1, 0, 0, 1, 0, ~
## $ parasite_load    <dbl> 28, 3, 12, 19, 15, 7, 38, 26, 6, 17, 10, 4, 5~
## $ respiratory_signs <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ injury_status    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
## $ cortisol_ng_ml   <dbl> 19.158612, 22.606257, 28.816896, 9.639142, 10~
## $ testosterone_ng_ml <dbl> 5.717721, 3.666248, 3.084256, 4.783930, 3.218~
```

```
# Summary of each column
summary(data)
```

```
## sample_id      gorilla_id      gorilla_name      group
## Length:240      Length:240      Length:240      Length:240
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## sample_date      observer      sex      age_class
## Min. :2025-01-02 Length:240      Length:240      Length:240
## 1st Qu.:2025-04-09 Class :character Class :character Class :character
## Median :2025-07-22 Mode :character Mode :character Mode :character
## Mean :2025-07-15
## 3rd Qu.:2025-10-18
## Max. :2025-12-31
##
## age_years      dominance_rank      group_size      fruit_availability_index
## Min. : 3.40      Min. :1.000      Min. :1.000      Min. :0.1256
## 1st Qu.:16.10      1st Qu.:2.000      1st Qu.:2.000      1st Qu.:0.4116
## Median :25.75      Median :3.000      Median :3.000      Median :0.5866
## Mean :24.22      Mean :3.889      Mean :3.467      Mean :0.5602
## 3rd Qu.:32.60      3rd Qu.:6.000      3rd Qu.:3.000      3rd Qu.:0.6832
## Max. :37.70      Max. :8.000      Max. :7.000      Max. :0.9500
## NA's :24
## feeding_time_prop aggression_events parasite_load      respiratory_signs
## Min. :0.2032      Min. :0.000      Min. : 0.00      Min. :0.0000
## 1st Qu.:0.3938      1st Qu.:0.000      1st Qu.: 9.00      1st Qu.:0.0000
## Median :0.5024      Median :1.000      Median :16.00      Median :0.0000
## Mean :0.5151      Mean :1.688      Mean :19.44      Mean :0.1333
```

```
## 3rd Qu.:0.6219      3rd Qu.:2.000      3rd Qu.: 26.00      3rd Qu.:0.0000
## Max.      :0.9229      Max.      :8.000      Max.      :111.00      Max.      :1.0000
##
## injury_status      cortisol_ng_ml      testosterone_ng_ml
## Min.      :0.00000      Min.      : 6.985      Min.      : 1.526
## 1st Qu.:0.00000      1st Qu.:12.570      1st Qu.: 3.788
## Median :0.00000      Median :16.370      Median : 5.855
## Mean      :0.08333      Mean      :18.050      Mean      : 7.152
## 3rd Qu.:0.00000      3rd Qu.:21.590      3rd Qu.: 9.084
## Max.      :1.00000      Max.      :55.701      Max.      :34.719
##
```

```
# Number of rows
nrow(data)
```

```
## [1] 240
```

```
# Number of columns
ncol(data)
```

```
## [1] 19
```

```
# Column names
names(data)
```

```
## [1] "sample_id"          "gorilla_id"
## [3] "gorilla_name"       "group"
## [5] "sample_date"        "observer"
## [7] "sex"                "age_class"
## [9] "age_years"          "dominance_rank"
## [11] "group_size"         "fruit_availability_index"
## [13] "feeding_time_prop"  "aggression_events"
## [15] "parasite_load"      "respiratory_signs"
## [17] "injury_status"      "cortisol_ng_ml"
## [19] "testosterone_ng_ml"
```

```
# =====
# Q: How many rows are in this dataset?
# Q: How many columns are in this dataset?
# Q: What does one row represent?
# =====
```

3. Identify Variable Types

A major goal this week is to identify different variable types.

Different variable types often require different statistical models.

3.1 Continuous Variables

A **continuous variable** is numeric and can take many possible values.

Examples in this dataset include:

- cortisol_ng_ml
- testosterone_ng_ml
- age_years
- fruit_availability_index

Let's look at some variables

```
summary(data$age_years)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      3.40   16.10   25.75   24.22   32.60   37.70
```

```
summary(data$sex)
```

```
##      Length      Class      Mode
##      240 character character
```

```
summary(data$parasite_load)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00    9.00   16.00   19.44   26.00   111.00
```

```
# =====
# Q: Which of these variables are continuous?
# =====
```

Check class

```
class(data$cortisol_ng_ml) # this is a numeric variable
```

```
## [1] "numeric"
```

as opposed to:

```
class(data$sex) # this is a character variable... which we can later convert to categorical
```

```
## [1] "character"
```

```
class(data$testosterone_ng_ml) # numeric
```

```
## [1] "numeric"
```

```
class(data$age_years) # numeric
```

```
## [1] "numeric"
```

```
# =====
# Q: What is the range of cortisol values?
# Q: What is the mean testosterone value?
# =====
```

3.2 Count Variables

A **count variable** is numeric, but it represents the number of times something happened.

Count variables usually:

- are whole numbers
- cannot be negative
- often have many zeros
- are often not normally distributed

Examples in this dataset include:

- aggression_events
- parasite_load

```
summary(data$aggression_events)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.000   0.000   1.000   1.688   2.000   8.000
```

```
summary(data$parasite_load)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     0.00   9.00   16.00   19.44   26.00  111.00
```

```
class(data$aggression_events) # class is numeric, and these are continuous, but they are counts and ten
```

```
## [1] "numeric"
```

```
class(data$parasite_load)
```

```
## [1] "numeric"
```

```
# note how the min and max are round numbers- 0 and 9 or 106
# the mean is a number with a decimal, because a mean of counts is not necessarily a round number. For
# mean(c(1,4,5))
```

```
# How many unique values?
```

```
unique(data$aggression_events)
```

```
## [1] 2 0 1 3 6 4 8 5 7
```

```
# Count how many rows have each aggression value
```

```
count(data, aggression_events)
```

```
## # A tibble: 9 x 2
##   aggression_events    n
##             <dbl> <int>
## 1                 0    73
## 2                 1    63
## 3                 2    45
## 4                 3    21
## 5                 4    17
## 6                 5    10
## 7                 6     7
## 8                 7     3
## 9                 8     1
```

what does this table tell you? What does each row mean?

```
# =====
# Q: Why is aggression_events a count variable rather than a continuous variable?
# Q: Can a count variable have decimal values?
# Q: Can a count variable be negative?
# =====
```

3.3 Binary Variables

A **binary variable** has two possible values.

Examples in this dataset include:

- `respiratory_signs`
- `injury_status`
- `sex` ...although `sex` might also have more categories, like *Unknown*, which will make it categorical, not binary!

In this dataset, `respiratory_signs` and `injury_status` are coded as:

- 0 = no
- 1 = yes

```
class(data$respiratory_signs)
```

```
## [1] "numeric"
```

```
class(data$injury_status)
```

```
## [1] "numeric"
```

```
class(data$sex)
```

```
## [1] "character"
```



```
table(data$respiratory_signs)
```

```
##  
##    0    1  
## 208   32
```

```
table(data$injury_status)
```

```
##  
##    0    1  
## 220   20
```

```
table(data$sex)
```

```
##  
##    F    M  
## 136  104
```

```
# =====  
# Q: What proportion of samples have respiratory signs?  
# Q: What proportion of samples have injury signs?  
# Q: Is sex binary in this dataset?  
# =====
```

```
# Proportion of samples with respiratory signs  
mean(data$respiratory_signs)
```

```
## [1] 0.1333333
```

```
# Proportion of samples with injury signs  
mean(data$injury_status)
```

```
## [1] 0.08333333
```

3.4 Categorical Variables

A **categorical variable** describes group membership.

Examples in this dataset include:

- group
- sex
- age_class
- observer

```
class(data$group)
```

```
## [1] "character"
```

```
class(data$age_class)
```

```
## [1] "character"
```

```
class(data$observer)
```

```
## [1] "character"
```

```
# Unique categories  
unique(data$group)
```

```
## [1] "BWE"      "Group5"   "MAF"      "SusaGp"   "PAB"      "ISA"  
## [7] "Group4"   "MSK"      "TOB"      "BEE"      "SHI"      "NunkieGp"
```

```
unique(data$age_class)
```

```
## [1] "adult"      "subadult"  "infant"
```

```
unique(data$observer)
```

```
## [1] "Mark"      "Sarah"     "Mary"      "Robert"    "Paul"
```

```
# Counts per category  
count(data, group, sort = TRUE)
```

```
## # A tibble: 12 x 2  
##   group      n  
##   <chr>    <int>  
## 1 PAB        56  
## 2 BEE        24  
## 3 Group4     24  
## 4 Group5     24  
## 5 SusaGp     24  
## 6 ISA        16  
## 7 MAF        16  
## 8 SHI        16  
## 9 TOB        16  
## 10 BWE        8  
## 11 MSK        8  
## 12 NunkieGp    8
```

```
count(data, age_class)
```

```
## # A tibble: 3 x 2  
##   age_class      n  
##   <chr>    <int>  
## 1 adult     216  
## 2 infant      8  
## 3 subadult   16
```

```
count(data, observer)
```

```
## # A tibble: 5 x 2
##   observer      n
##   <chr>    <int>
## 1 Mark      49
## 2 Mary      48
## 3 Paul      58
## 4 Robert    39
## 5 Sarah     46
```

```
# =====
# Q: Which group has the most observations?
# Q: Which age class has the most observations?
# Q: Why is observer a categorical variable?
# =====
```

3.5 Proportion Variables

A **proportion variable** is numeric but bounded between 0 and 1.

In this dataset, `feeding_time_prop` represents the proportion of observed time spent feeding.

```
class(data$feeding_time_prop)
```

```
## [1] "numeric"
```

```
summary(data$feeding_time_prop)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2032  0.3938   0.5024  0.5151  0.6219  0.9229
```

```
range(data$feeding_time_prop)
```

```
## [1] 0.2032042 0.9229194
```

```
# =====
# Q: Why is feeding_time_prop not a count variable?
# Q: What are the lowest and highest feeding time proportions?
# Q: Why might proportions require special attention in statistical models?
# =====
```

4. Cross-Sectional vs. Longitudinal Data

A **cross-sectional** dataset has one observation per individual or unit.

A **longitudinal** dataset follows the same individuals over time.

This dataset contains repeated samples from the same gorillas, so it has a longitudinal or repeated-measures structure.

```
# How many samples per gorilla?
samples_per_gorilla <- data %>%
  count(gorilla_id, gorilla_name, sort = TRUE)

samples_per_gorilla
```

```
## # A tibble: 30 x 3
##   gorilla_id gorilla_name     n
##   <chr>      <chr>         <int>
## 1 G001      Gasore             8
## 2 G002      Poppy              8
## 3 G003      Ishimwe            8
## 4 G004      Umuyaga            8
## 5 G005      Ubusugire          8
## 6 G006      Mushya             8
## 7 G007      Chundura           8
## 8 G008      Twihangane         8
## 9 G009      Gitebe             8
## 10 G010     Petula             8
## # i 20 more rows
```

```
# How many unique gorillas?
length(unique(data$gorilla_id))
```

```
## [1] 30
```

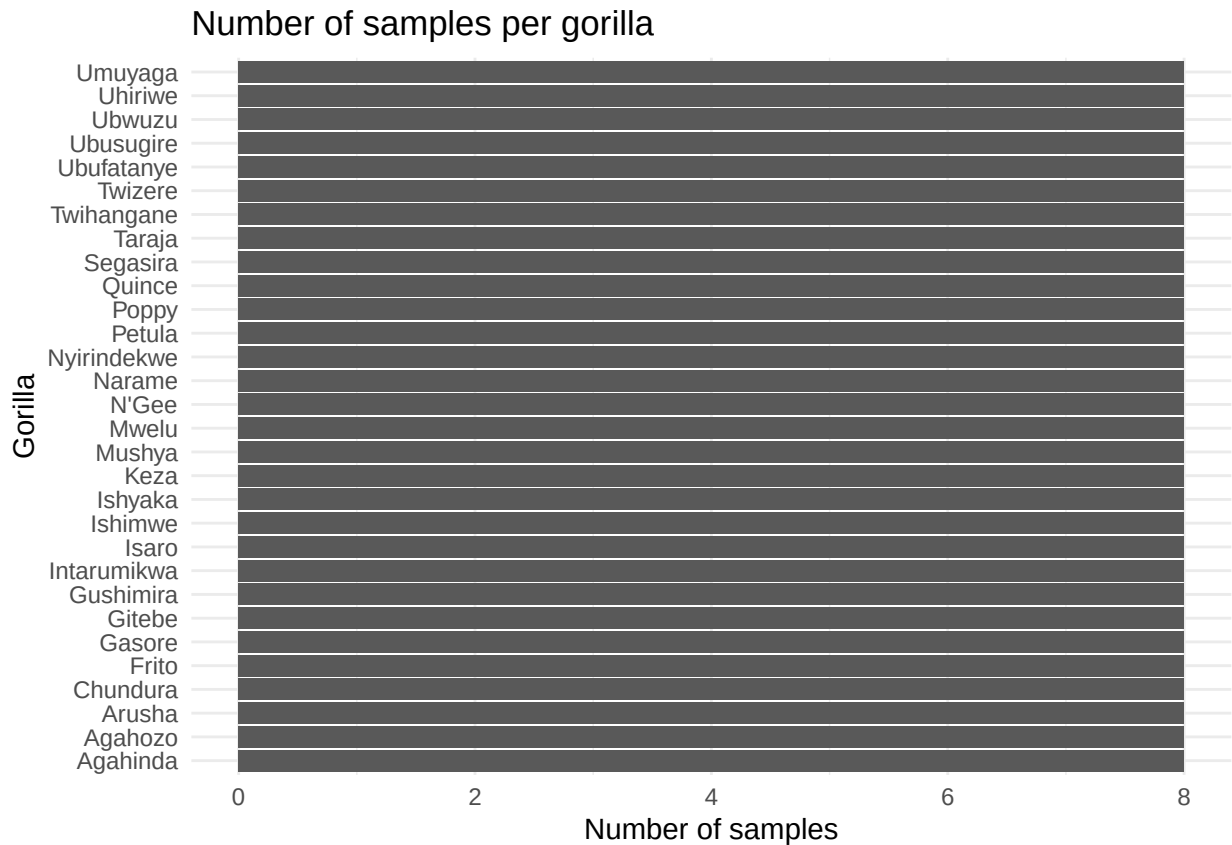
```
# How many total samples?
nrow(data)
```

```
## [1] 240
```

```
# =====
# Q: Do all gorillas have more than one sample?
# Q: Why does repeated sampling matter for statistical models?
# Q: Are these observations independent?
# =====
```

4.1 Visualize repeated measures

```
ggplot(samples_per_gorilla, aes(x = reorder(gorilla_name, n), y = n)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Number of samples per gorilla",
    x = "Gorilla",
    y = "Number of samples"
  ) +
  theme_minimal()
```



If the same individual appears multiple times, observations from that individual are likely more similar to each other than to observations from other individuals. This violates the assumption of full independence and is one reason we may later use mixed-effects models.

5. Diagnose Distributions

Before choosing a model, we need to look at the distribution of the response variable.

We will practice with:

- histograms
- density plots
- QQ plots

5.1 Histogram with `geom_histogram()`

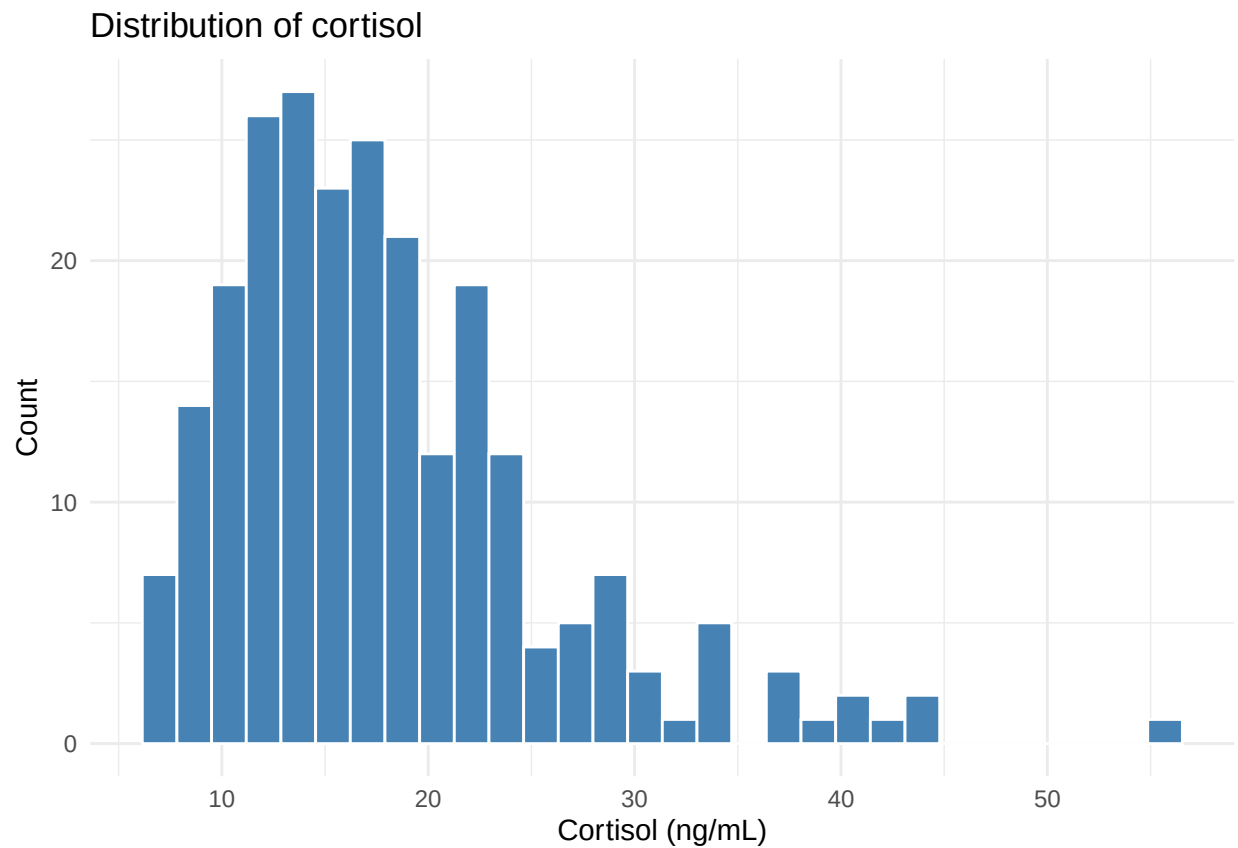
A histogram shows how often values fall within different ranges.

```
ggplot(data, aes(x = cortisol_ng_ml)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(
    title = "Distribution of cortisol",
    x = "Cortisol (ng/mL)",
```

```

y = "Count"
) +
theme_minimal()

```



```

# =====
# Q: Is cortisol evenly distributed?
# Q: Does it look normally distributed?
# Q: Is it right-skewed, left-skewed, or roughly symmetric?
# =====

```

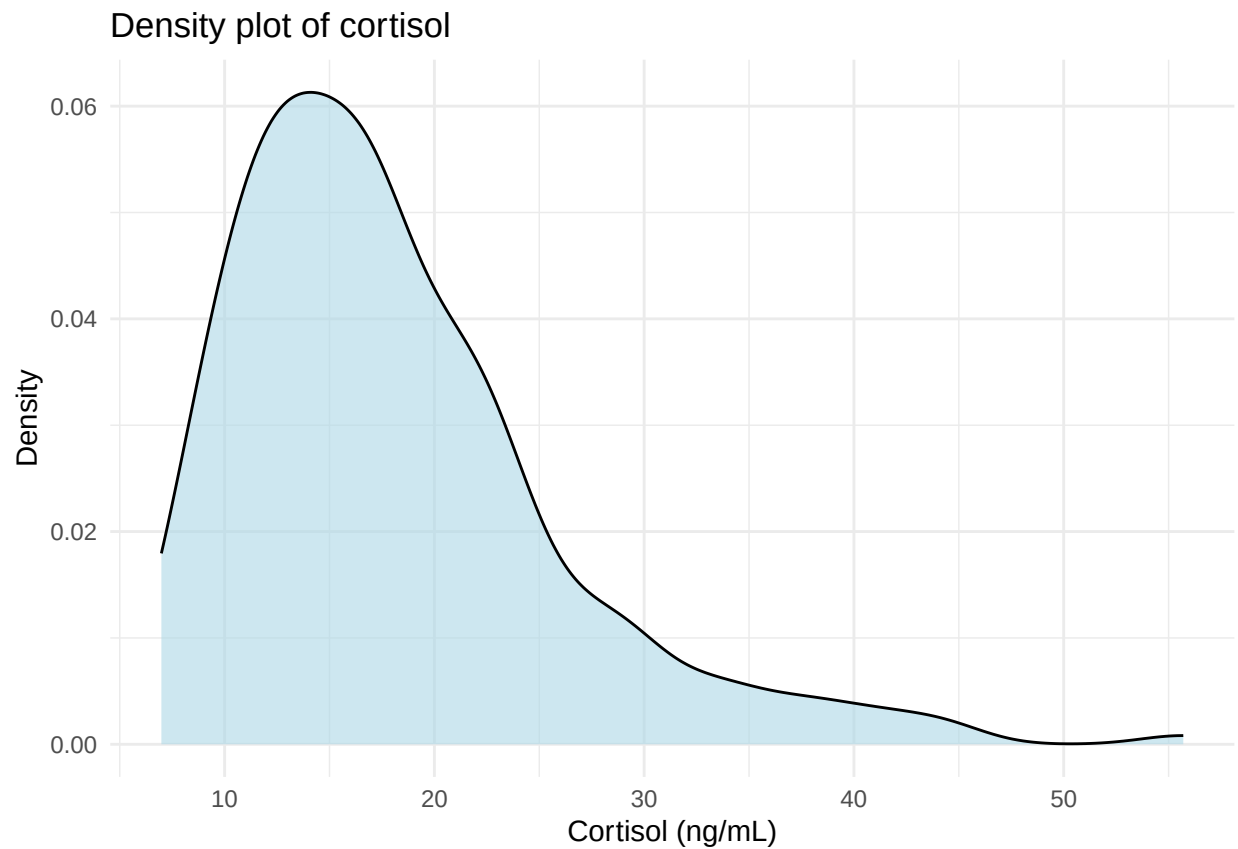
5.2 Density plot with geom_density()

A density plot shows a smoothed version of the distribution.

```

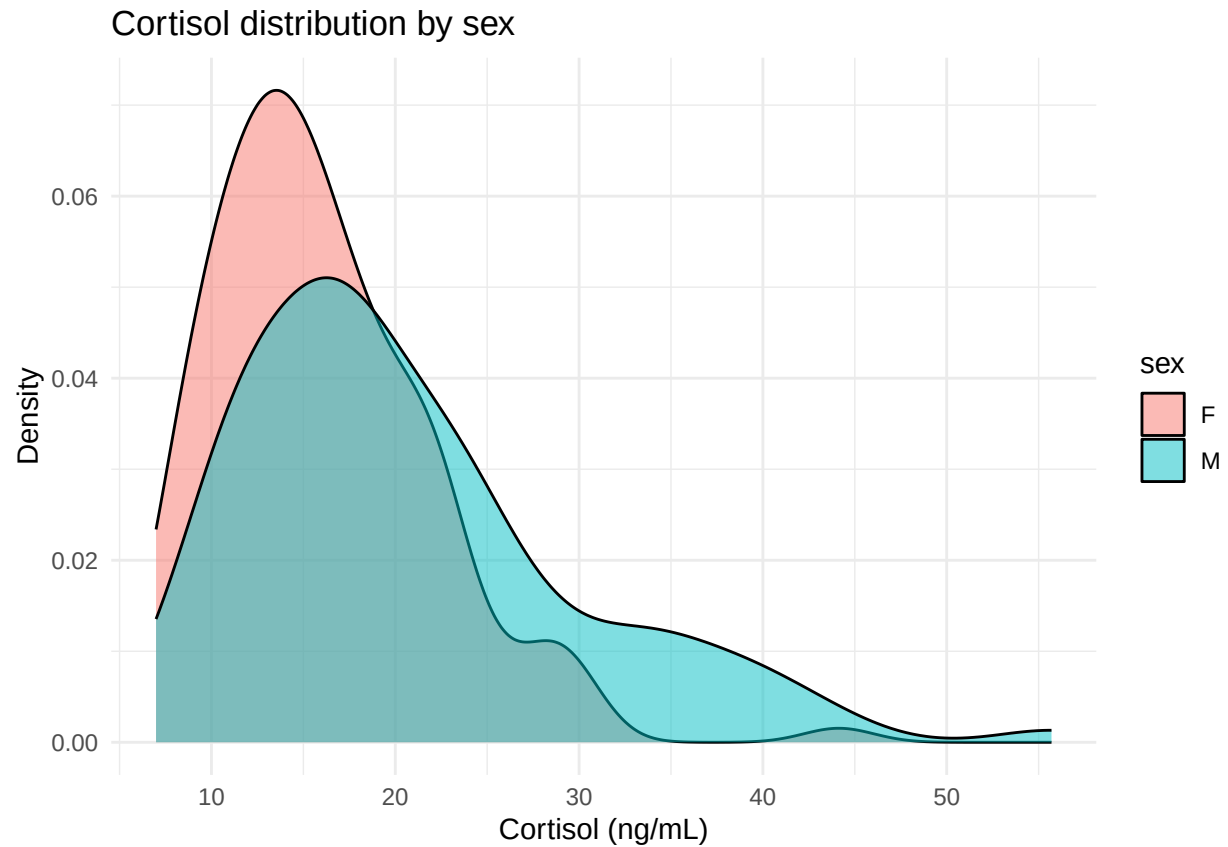
ggplot(data, aes(x = cortisol_ng_ml)) +
  geom_density(fill = "lightblue", alpha = 0.6) +
  labs(
    title = "Density plot of cortisol",
    x = "Cortisol (ng/mL)",
    y = "Density"
  ) +
  theme_minimal()

```



5.3 Compare distributions by group

```
ggplot(data, aes(x = cortisol_ng_ml, fill = sex)) +  
  geom_density(alpha = 0.5) +  
  labs(  
    title = "Cortisol distribution by sex",  
    x = "Cortisol (ng/mL)",  
    y = "Density"  
  ) +  
  theme_minimal()
```



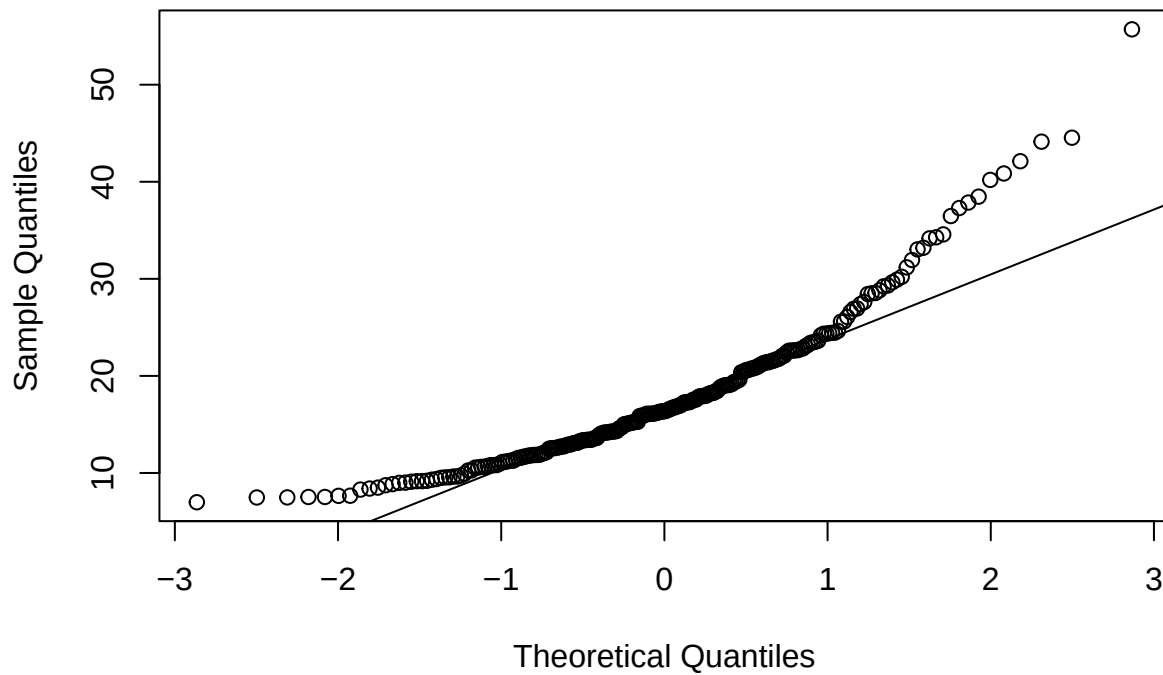
5.4 QQ plot with `qqnorm()` and `qqline()`

QQ plots help us visually compare our data to a normal distribution.

If the points fall close to the line, the variable is closer to normally distributed.

```
qqnorm(data$cortisol_ng_ml)
qqline(data$cortisol_ng_ml)
```


Normal Q-Q Plot



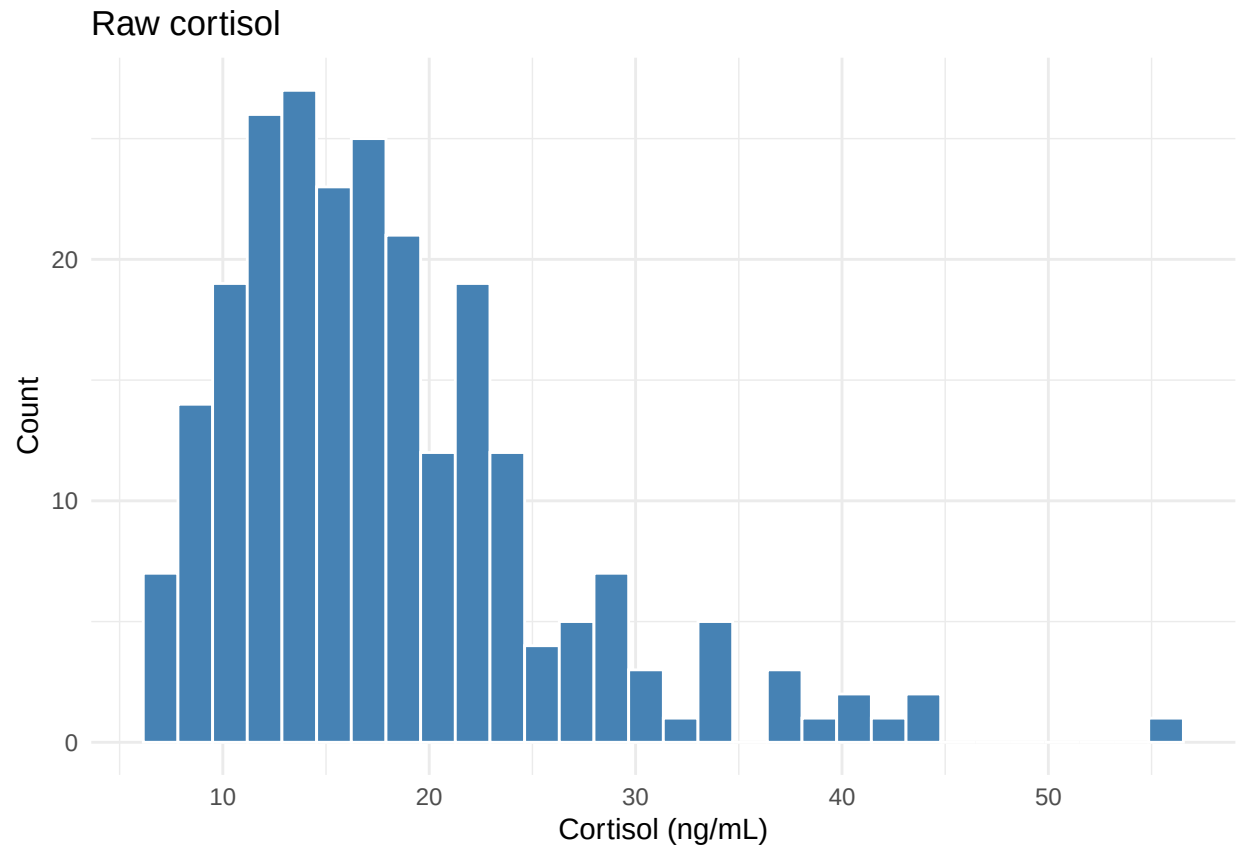
```
# =====
# Q: Do the cortisol values follow the line closely?
# Q: What does this tell us about whether cortisol is normally distributed?
# =====
```

5.5 Compare raw and log-transformed cortisol

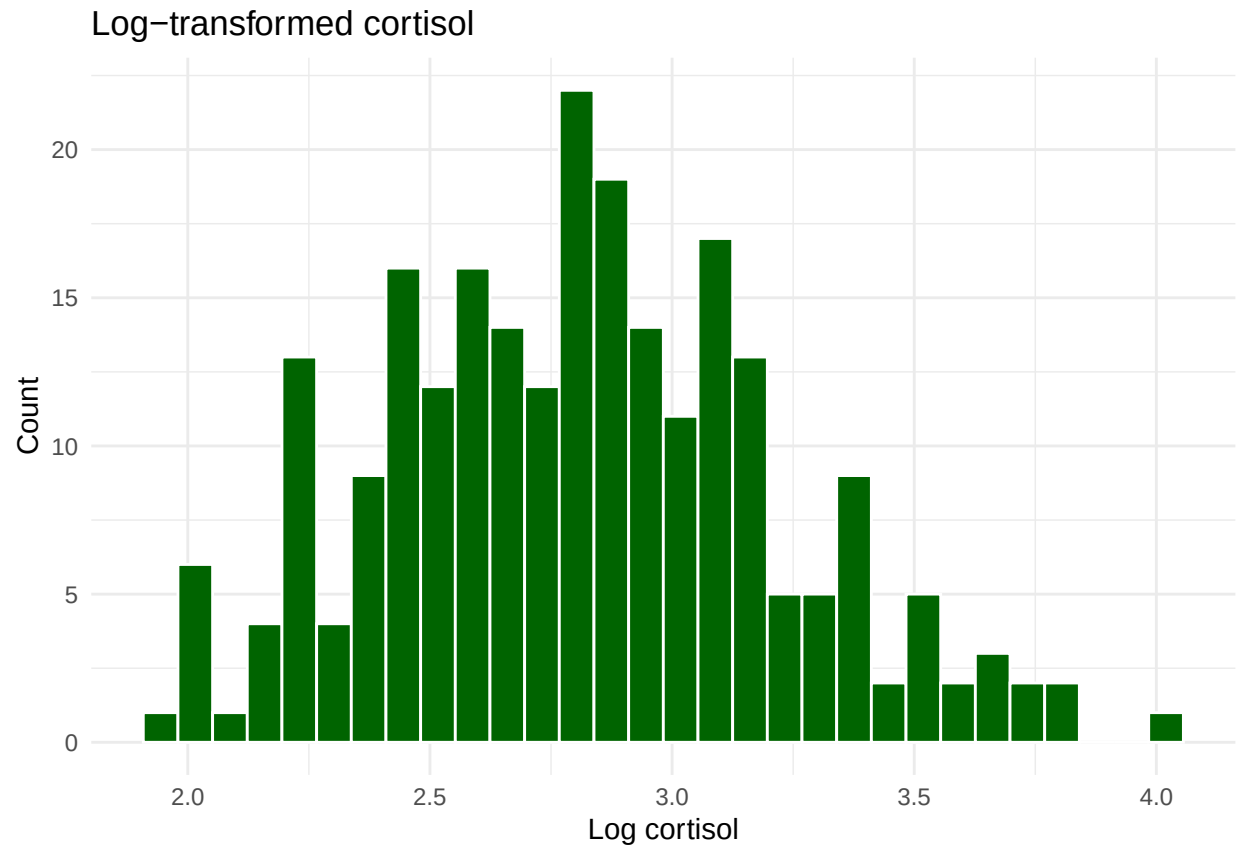
Hormone data are often right-skewed. A log transformation can sometimes make the distribution more symmetric.

```
data <- data %>%
  mutate(
    log_cortisol_ng_ml = log(cortisol_ng_ml),
    log_testosterone_ng_ml = log(testosterone_ng_ml)
  )

# raw cortisol
ggplot(data, aes(x = cortisol_ng_ml)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(
    title = "Raw cortisol",
    x = "Cortisol (ng/mL)",
    y = "Count"
  ) +
  theme_minimal()
```



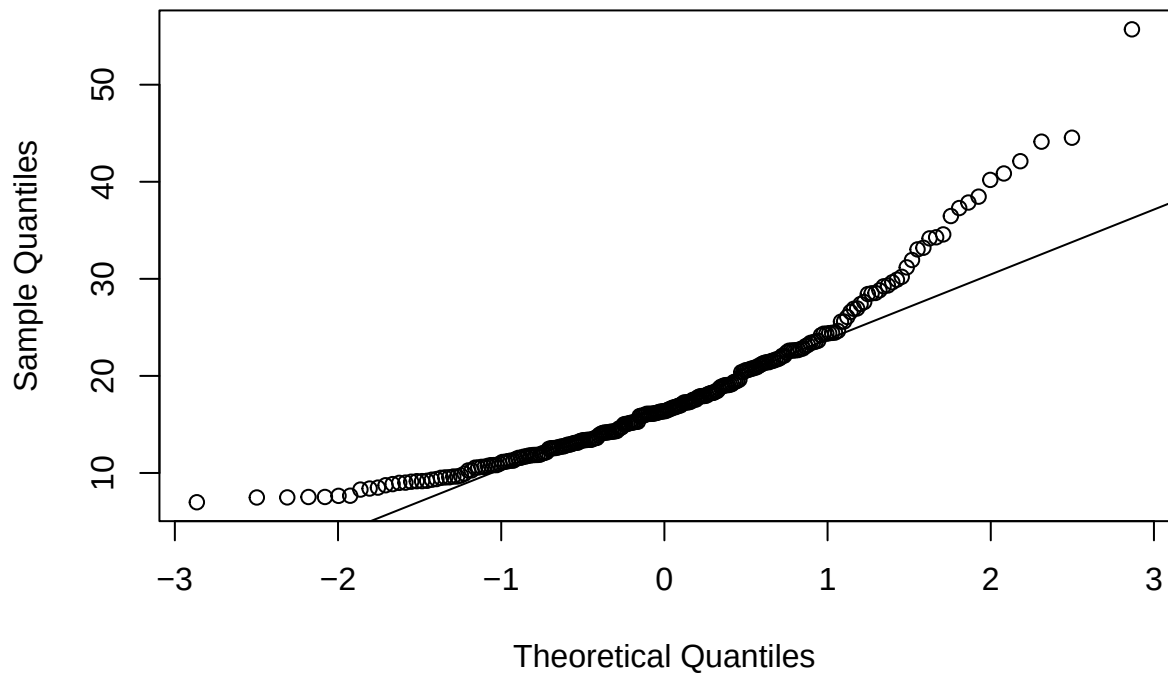
```
# log cortisol
ggplot(data, aes(x = log_cortisol_ng_ml)) +
  geom_histogram(bins = 30, fill = "darkgreen", color = "white") +
  labs(
    title = "Log-transformed cortisol",
    x = "Log cortisol",
    y = "Count"
  ) +
  theme_minimal()
```



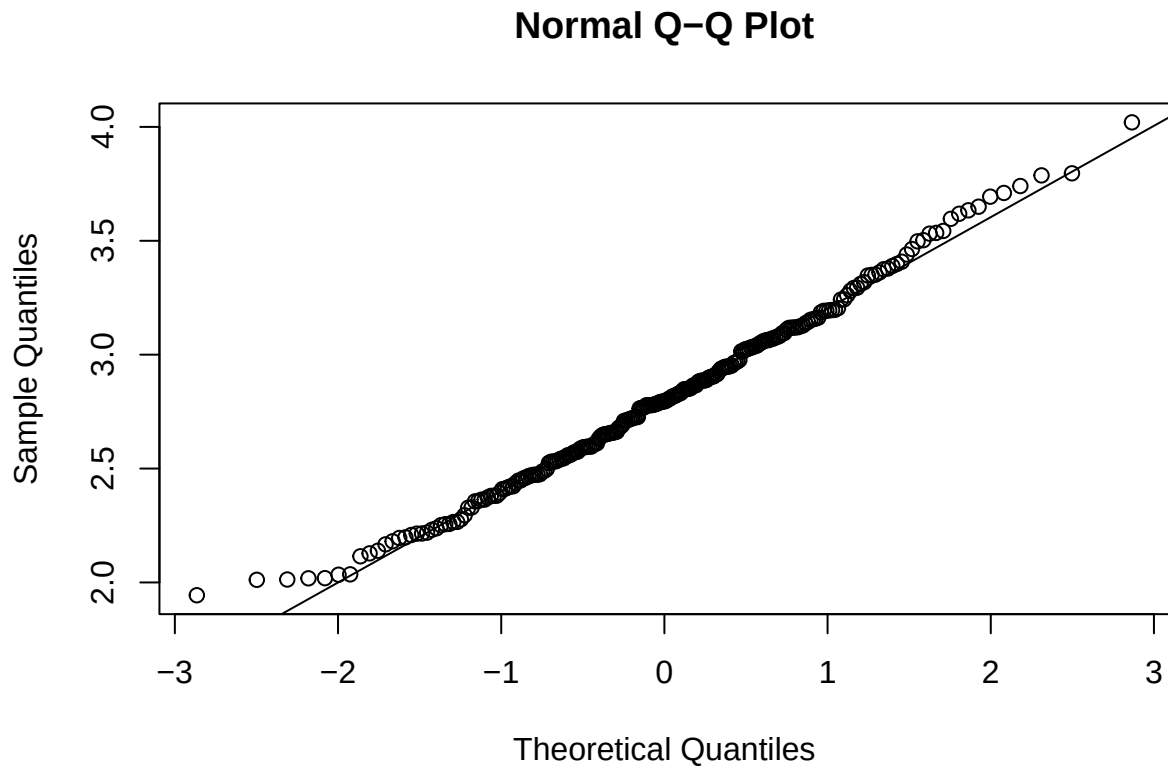
5.6 QQ plots for CORT

```
# QQ plot for raw cortisol  
qqnorm(data$cortisol_ng_ml)  
qqline(data$cortisol_ng_ml)
```

Normal Q-Q Plot



```
# QQ plot for log cortisol  
qqnorm(data$log_cortisol_ng_ml)  
qqline(data$log_cortisol_ng_ml)
```



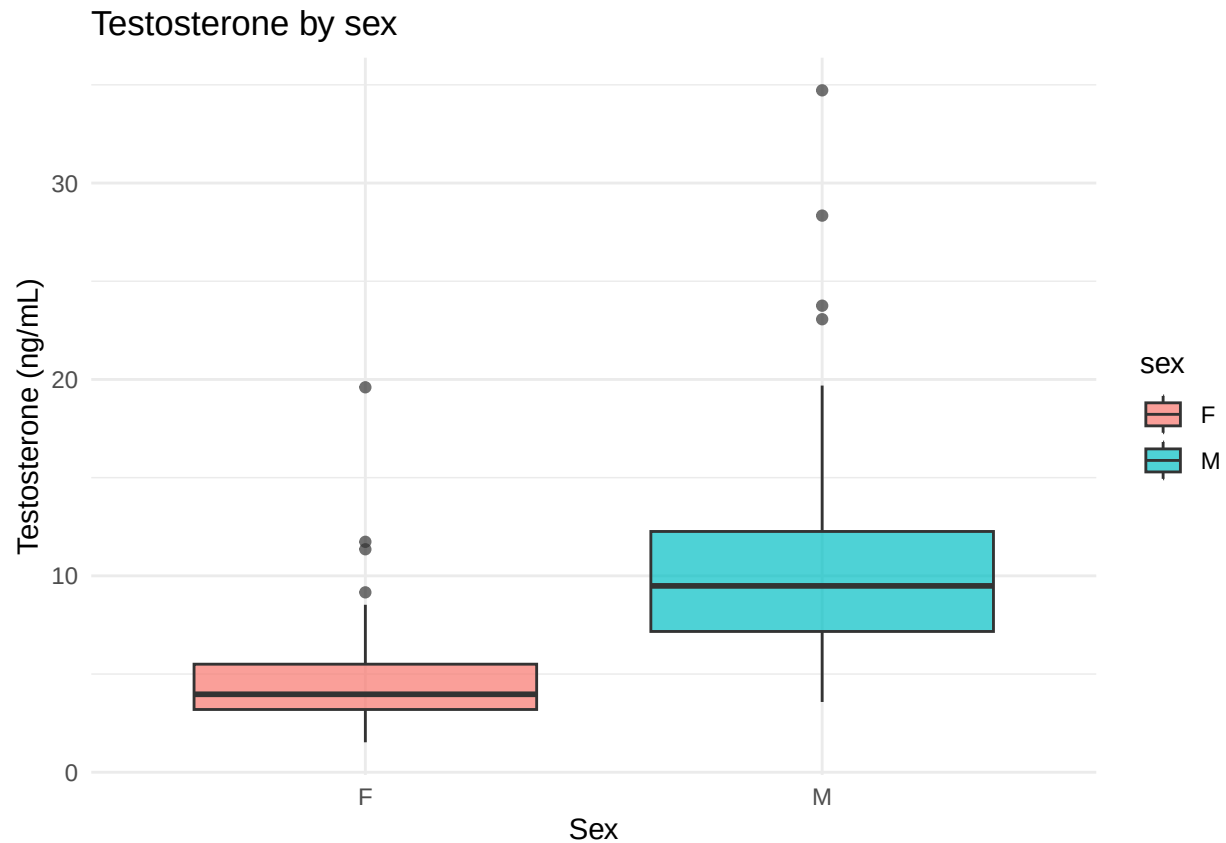
```
# =====
# Q: Which looks more normally distributed: raw cortisol or log cortisol?
# Q: Why might this matter for choosing a model?
# =====
```

6. Visualize Grouped Summaries

Grouped plots are useful for comparing values across categories.

6.1 Testosterone by sex

```
ggplot(data, aes(x = sex, y = testosterone_ng_ml, fill = sex)) +
  geom_boxplot(alpha = 0.7) +
  labs(
    title = "Testosterone by sex",
    x = "Sex",
    y = "Testosterone (ng/mL)"
  ) +
  theme_minimal()
```



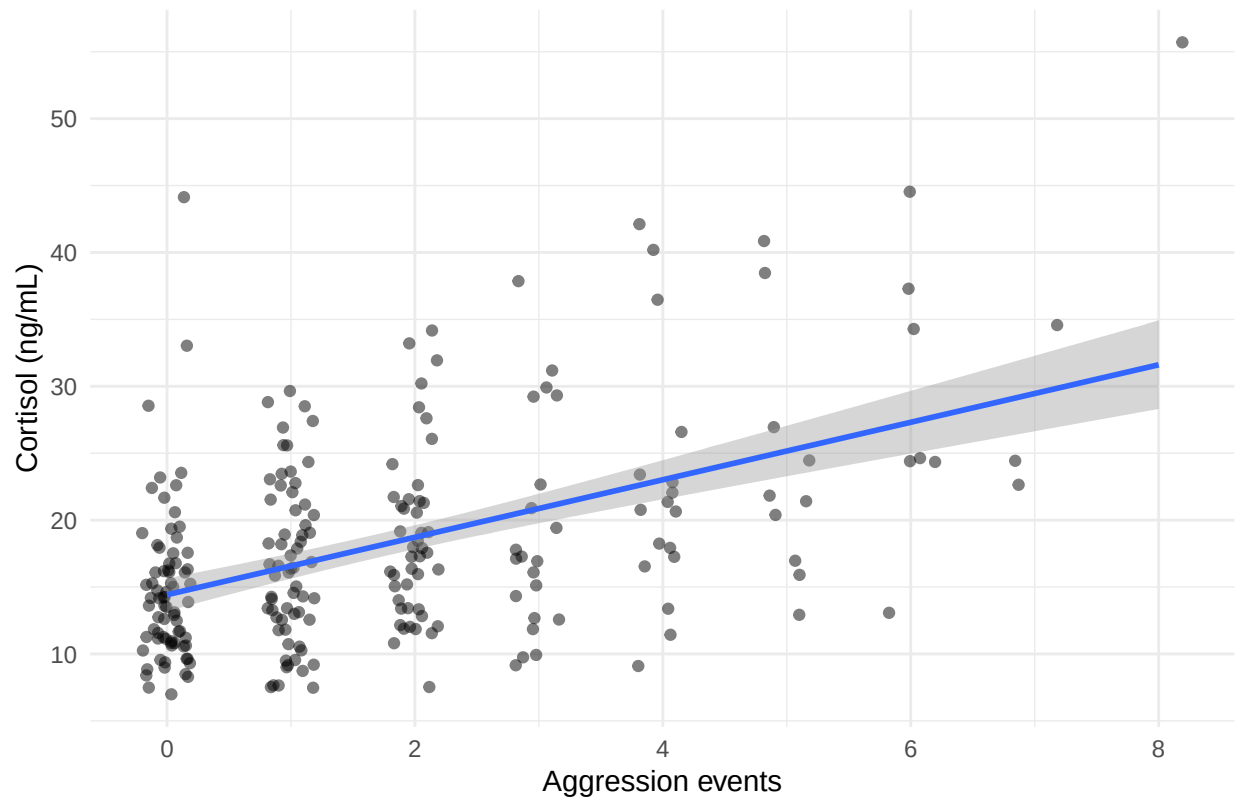
```
# =====
# Q: Which sex has higher testosterone values in this dataset?
# Q: Is there overlap between groups?
# Q: What kind of model might we use if testosterone were the response variable?
# =====
```

6.2 Cortisol and aggression

```
ggplot(data, aes(x = aggression_events, y = cortisol_ng_ml)) +
  geom_jitter(width = 0.2, alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Cortisol and aggression events",
    x = "Aggression events",
    y = "Cortisol (ng/mL)"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

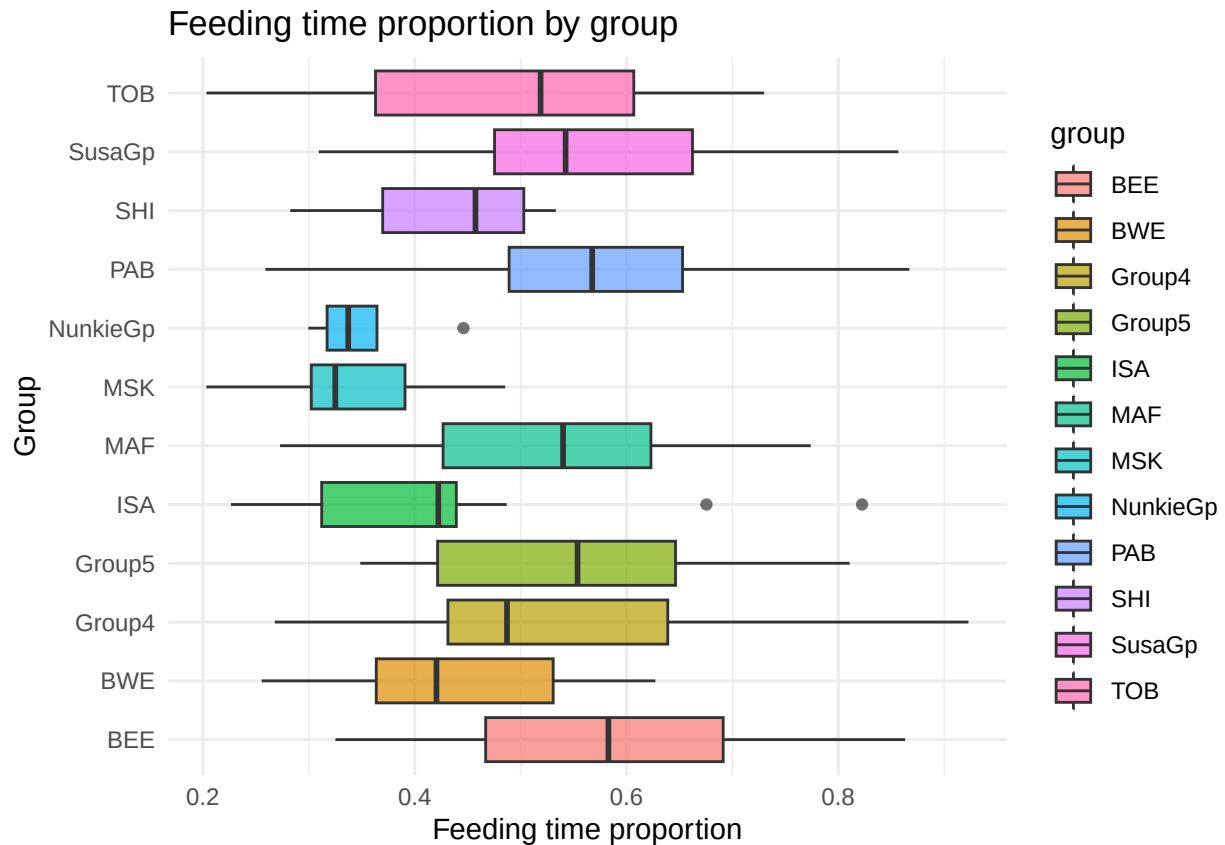
Cortisol and aggression events



```
# =====
# Q: What kind of variable is aggression_events?
# Q: What kind of variable is cortisol_ng_ml?
# Q: Does the relationship look positive, negative, or flat?
# =====
```

6.3 Feeding time by group

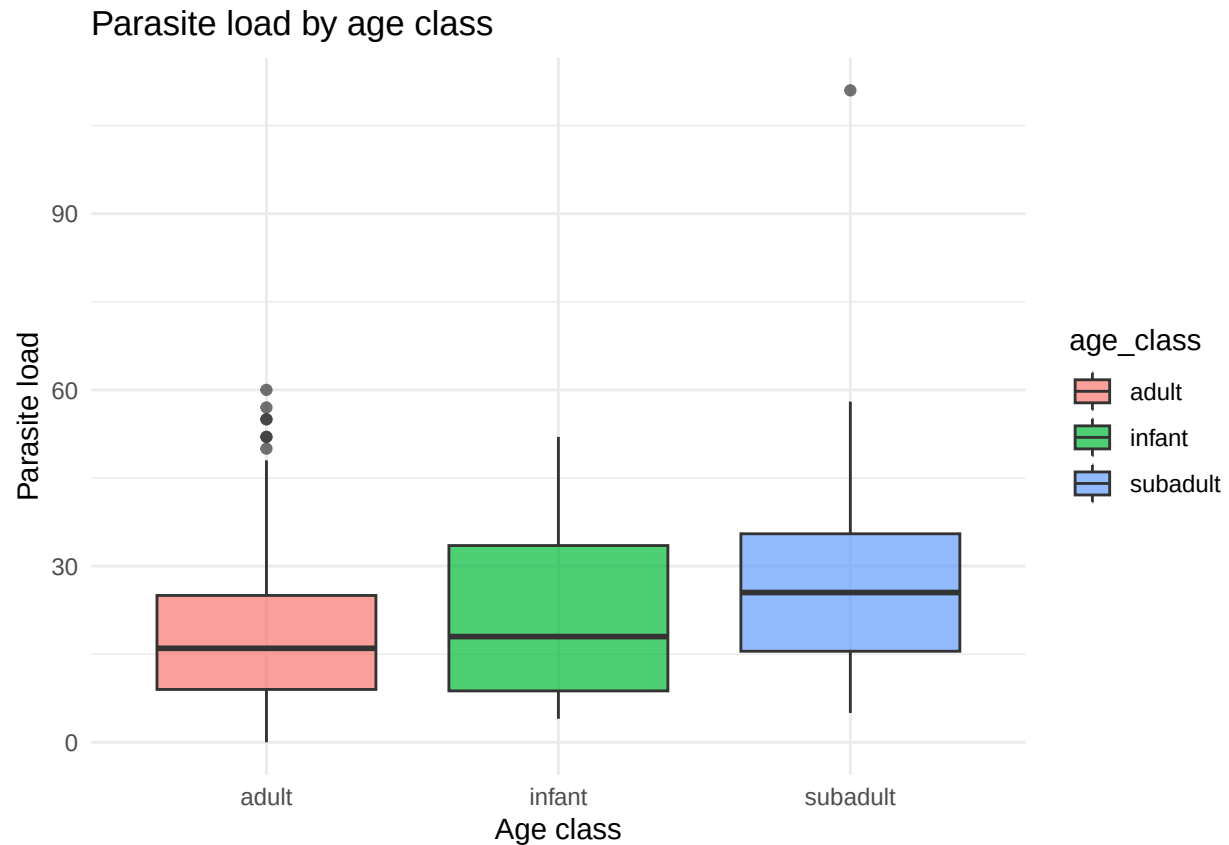
```
ggplot(data, aes(x = group, y = feeding_time_prop, fill = group)) +
  geom_boxplot(alpha = 0.7) +
  coord_flip() +
  labs(
    title = "Feeding time proportion by group",
    x = "Group",
    y = "Feeding time proportion"
  ) +
  theme_minimal()
```



```
# =====
# Q: What kind of variable is feeding_time_prop?
# Q: What kind of variable is group?
# Q: Why might group-level differences matter biologically?
# =====
```

6.4 Parasite load by age class

```
ggplot(data, aes(x = age_class, y = parasite_load, fill = age_class)) +
  geom_boxplot(alpha = 0.7) +
  labs(
    title = "Parasite load by age class",
    x = "Age class",
    y = "Parasite load"
  ) +
  theme_minimal()
```

```
# =====
# Q: What kind of variable is parasite_load?
# Q: What kind of variable is age_class?
# Q: Would a standard linear model be the best choice for parasite_load? Why or why not?
# =====
```

6.5 Respiratory signs by group size

For binary variables, we often summarize the proportion of observations coded as 1.

```
resp_summary <- data %>%
  group_by(group, group_size) %>%
  dplyr::summarise(
    respiratory_prop = mean(respiratory_signs),
    n = n(),
    .groups = "drop"
  )

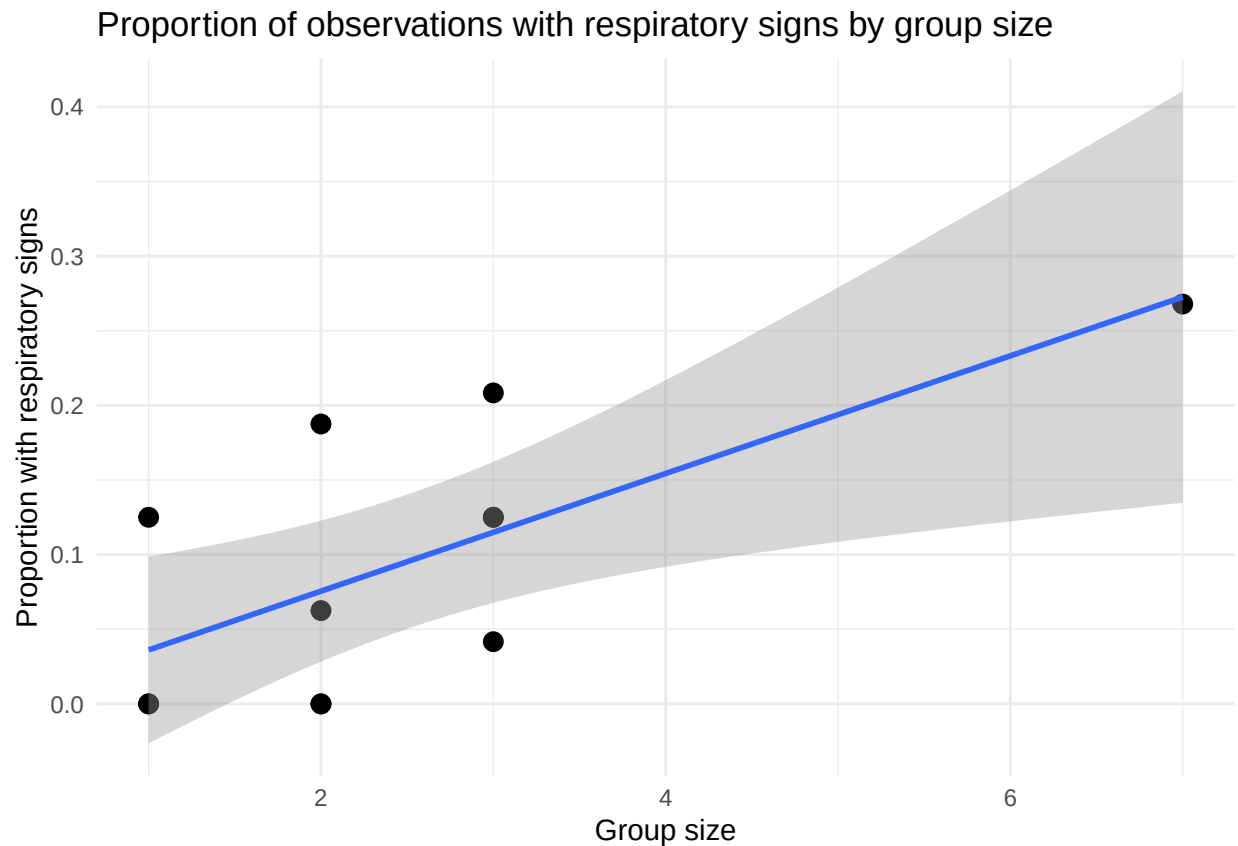
resp_summary
```

```
## # A tibble: 12 x 4
##   group   group_size respiratory_prop     n
##   <chr>       <dbl>         <dbl> <int>
## 1 BEE           3          0.208    24
```

```
## 2 BWE          1          0          8
## 3 Group4       3      0.0417      24
## 4 Group5       3      0.125      24
## 5 ISA          2      0.188      16
## 6 MAF          2          0      16
## 7 MSK          1      0.125       8
## 8 NunkieGp     1          0       8
## 9 PAB          7      0.268      56
## 10 SHI         2      0.0625     16
## 11 SusaGp      3      0.125      24
## 12 TOB         2          0      16
```

```
ggplot(resp_summary, aes(x = group_size, y = respiratory_prop)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm") +
  labs(
    title = "Proportion of observations with respiratory signs by group size",
    x = "Group size",
    y = "Proportion with respiratory signs"
  ) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
# =====
# Q: What kind of variable is respiratory_signs?
# Q: Why did we summarize it using mean()?
# Q: What does a mean of 0.25 mean for a binary 0/1 variable?
# =====
```

7. From Research Question to Model Type

A statistical model should follow from the biological question and the response variable.

The most important first question is:

What kind of response variable do I have?

7.1 Simple statistical decision framework

Biological question	Response variable	Response type	Possible model family
Does aggression predict cortisol?	<code>cortisol_ng_ml</code>	continuous, skewed	linear model on log-transformed response
Does sex predict testosterone?	<code>testosterone_ng_ml</code>	continuous, skewed	linear model on log-transformed response
Does age class predict parasite load?	<code>parasite_load</code>	count	Poisson or negative binomial model
Does group size predict respiratory signs?	<code>respiratory_signs</code>	binary	logistic regression
Does group predict feeding time?	<code>feeding_time_prop</code>	proportion	beta regression or transformed linear model
Do individuals differ in hormone values over time?	repeated samples	non-independent	mixed-effects model

7.2 Important model-selection questions

Before choosing a model, ask:

1. What is the response variable?
2. What type of variable is the response?
3. Is the response approximately normal, skewed, count, binary, or proportion?
4. Are observations independent?
5. Are individuals sampled repeatedly?
6. Is the dataset cross-sectional or longitudinal?
7. What are the predictor variables?
8. Are predictors continuous, categorical, binary, count, or proportion?

9. Does the research question involve group differences, relationships, prediction, or repeated measures?

7.3 Example 1: Continuous response

Research question:

Do gorillas with more aggression events have higher cortisol?

Response variable:

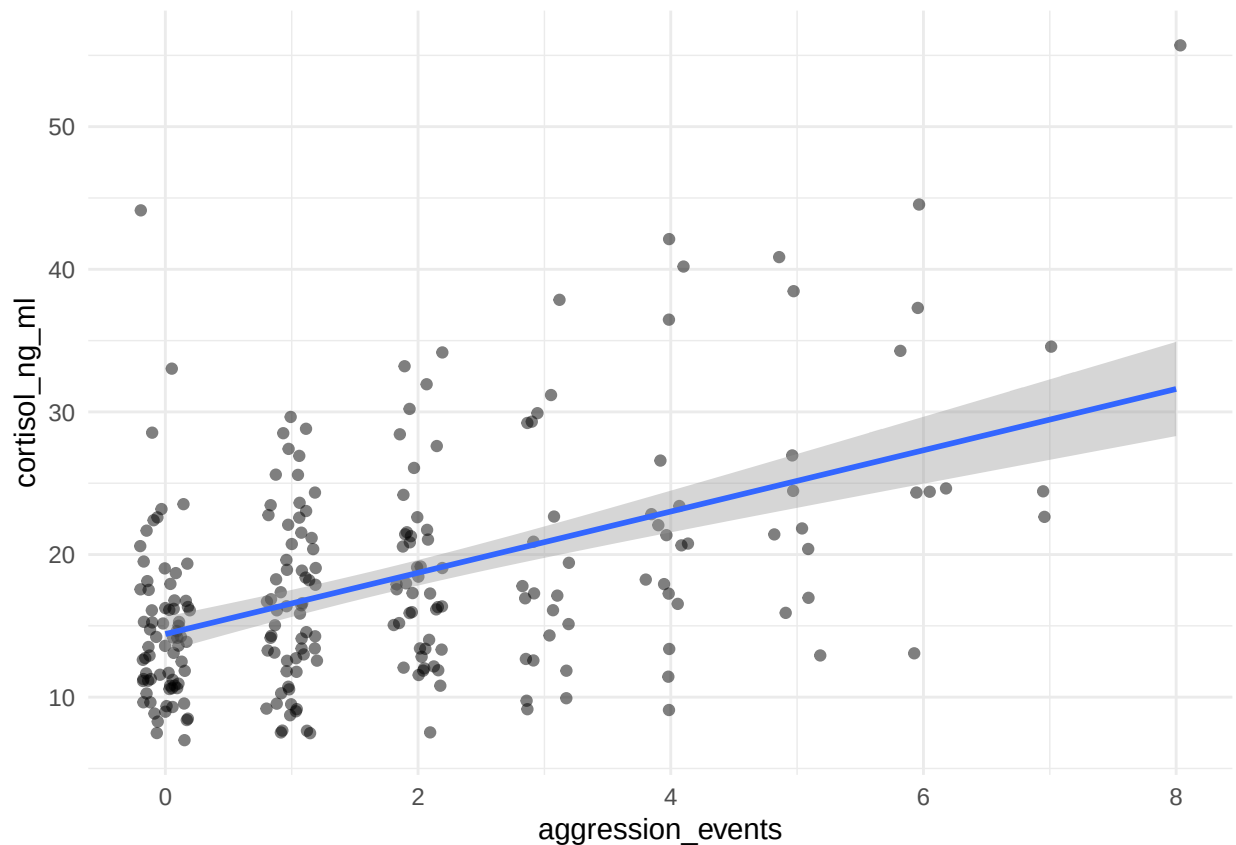
cortisol_ng_ml

Predictor variable:

aggression_events

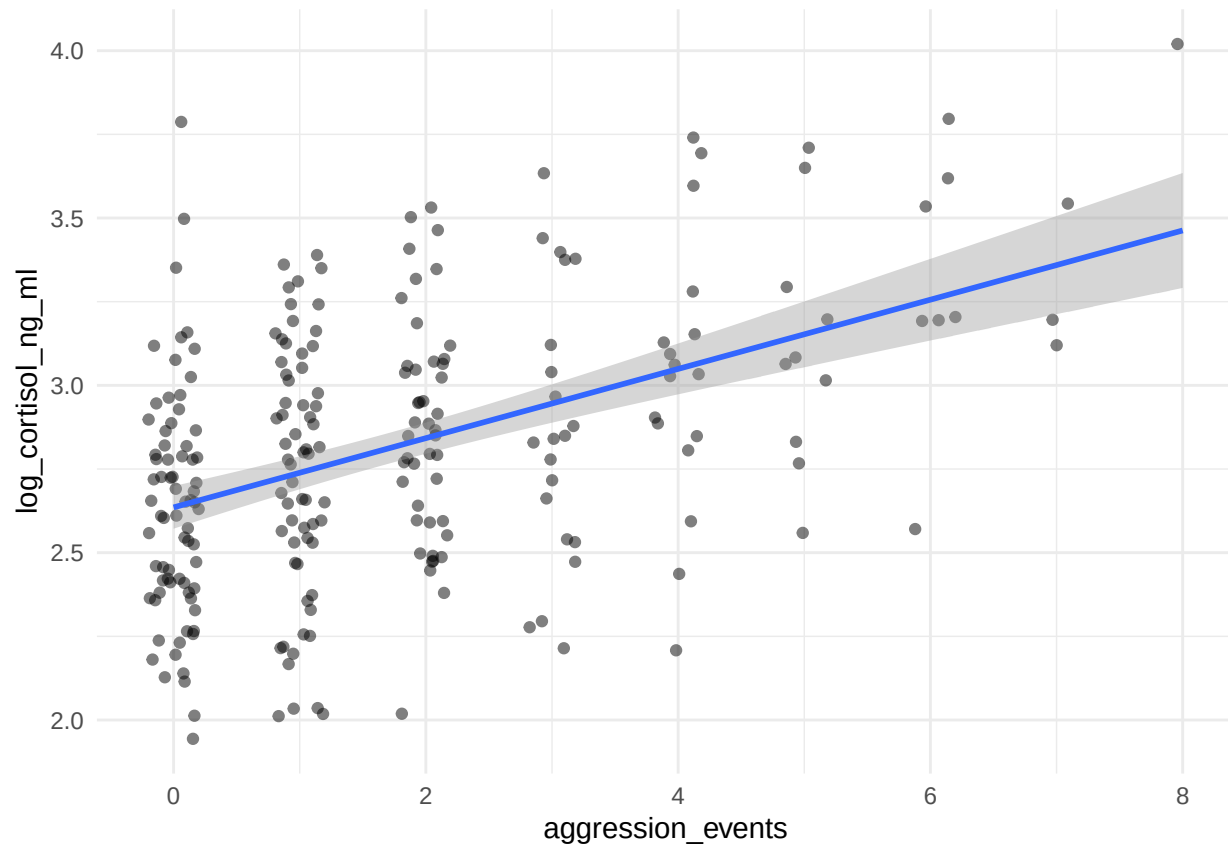
```
# Raw response
ggplot(data, aes(x = aggression_events, y = cortisol_ng_ml)) +
  geom_jitter(width = 0.2, alpha = 0.5) +
  geom_smooth(method = "lm") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
# Log-transformed response
ggplot(data, aes(x = aggression_events, y = log_cortisol_ng_ml)) +
  geom_jitter(width = 0.2, alpha = 0.5) +
  geom_smooth(method = "lm") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
# =====
# Q: What is the response variable?
# Q: What type of variable is the response?
# Q: Are observations independent?
# Q: What kind of model might be appropriate?
# =====
```

7.4 Example 2: Count response

Research question:

Does parasite load differ by age class?

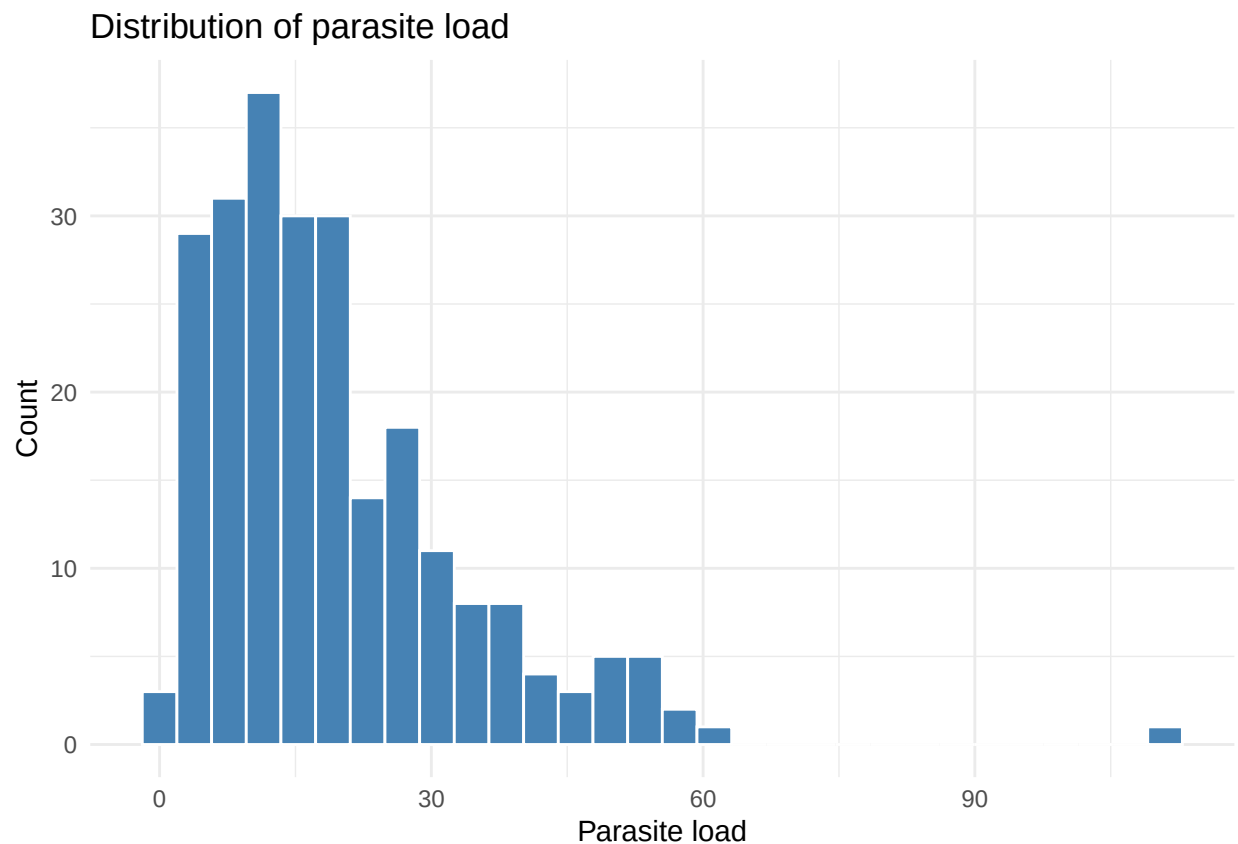
Response variable:

parasite_load

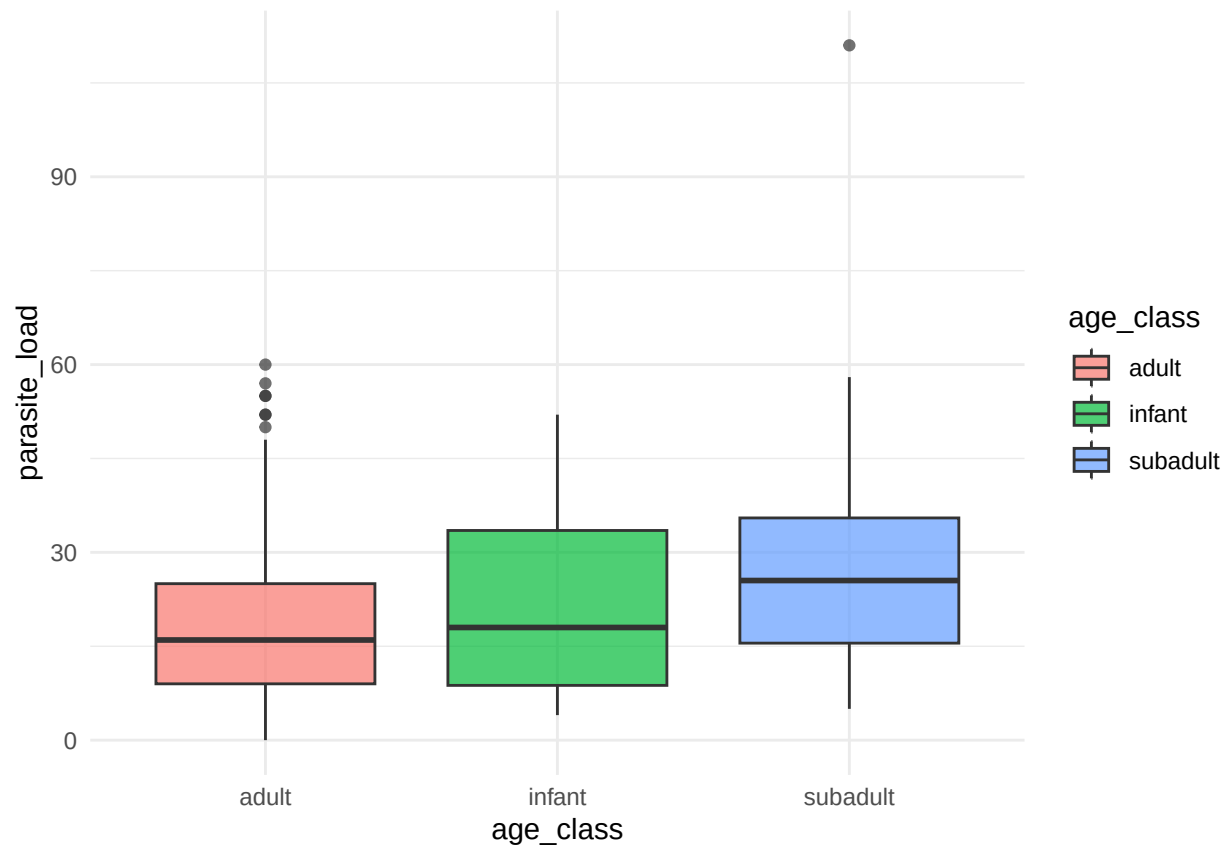
Predictor variable:

age_class

```
ggplot(data, aes(x = parasite_load)) +  
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +  
  labs(  
    title = "Distribution of parasite load",  
    x = "Parasite load",  
    y = "Count"  
  ) +  
  theme_minimal()
```



```
ggplot(data, aes(x = age_class, y = parasite_load, fill = age_class)) +  
  geom_boxplot(alpha = 0.7) +  
  theme_minimal()
```



```
# =====
# Q: What is the response variable?
# Q: What type of variable is the response?
# Q: Why might a normal linear model not be ideal here?
# Q: What kind of model might be appropriate?
# =====
```

7.5 Example 3: Binary response

Research question:

Are respiratory signs more likely in larger groups?

Response variable:

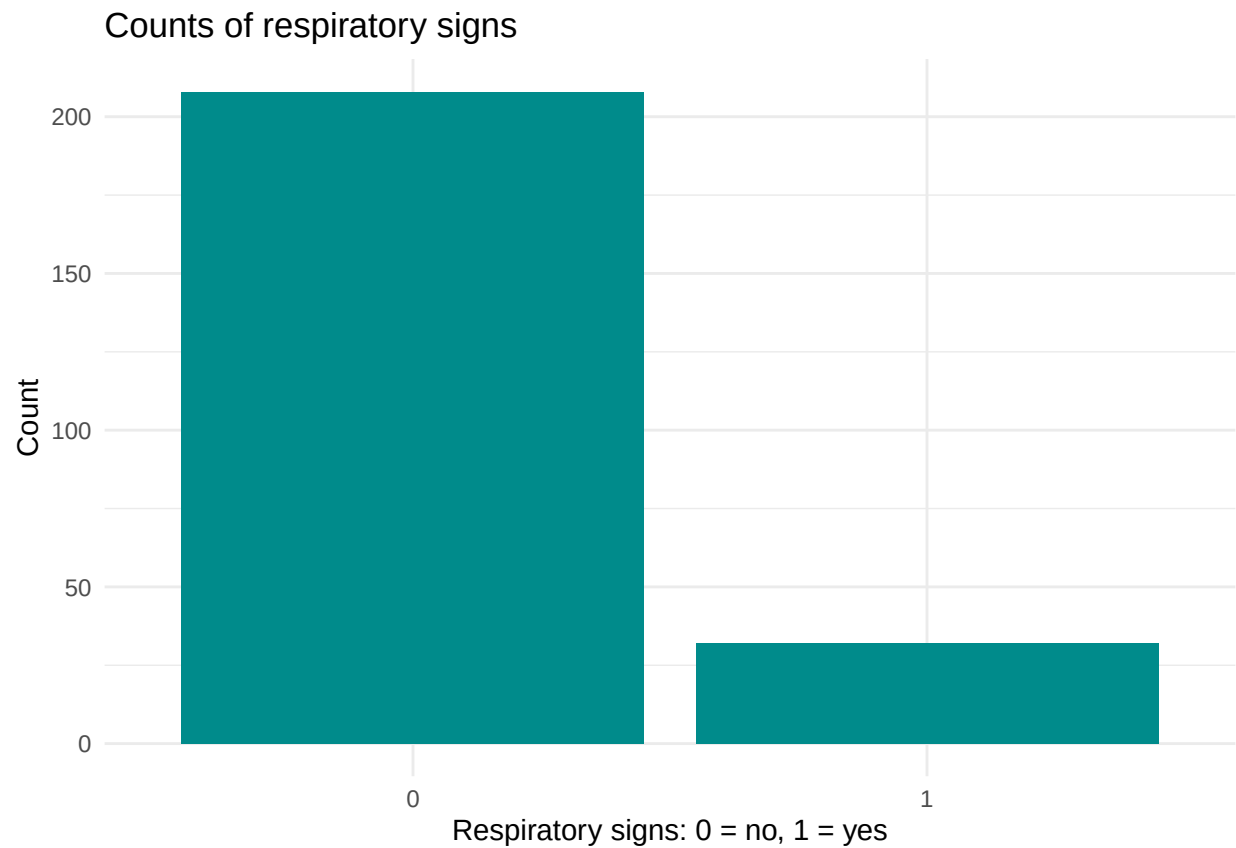
respiratory_signs

Predictor variable:

group_size

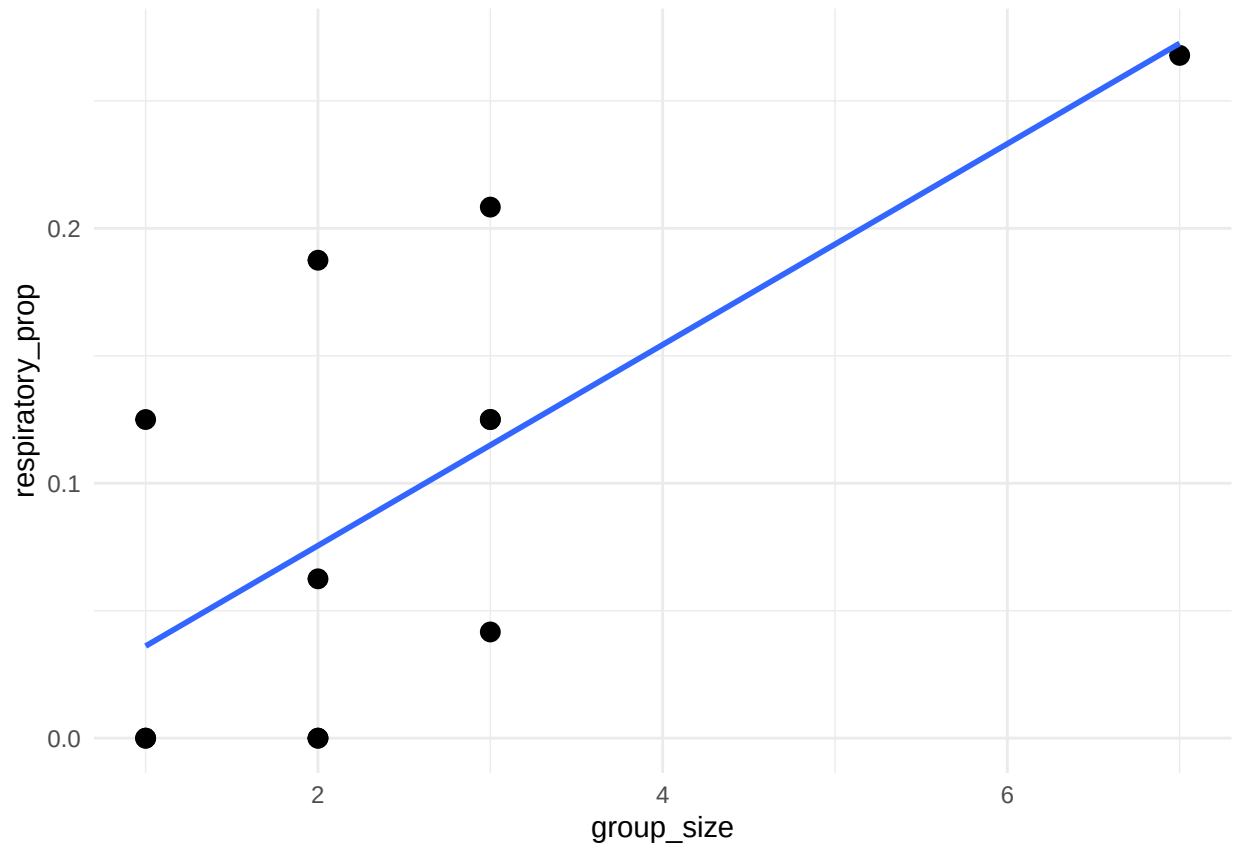
```
ggplot(data, aes(x = factor(respiratory_signs))) +
  geom_bar(fill = "cyan4") +
  labs(
    title = "Counts of respiratory signs",
    x = "Respiratory signs: 0 = no, 1 = yes",
```

```
y = "Count"
) +
theme_minimal()
```



```
ggplot(
  resp_summary,
  aes(x = group_size, y = respiratory_prop)
) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()
```

'geom_smooth()' using formula = 'y ~ x'



```
# =====
# Q: What is the response variable?
# Q: What type of variable is the response?
# Q: What kind of model might be appropriate?
# =====
```

7.6 Example 4: Proportion response

Research question:

Does feeding time proportion differ by group?

Response variable:

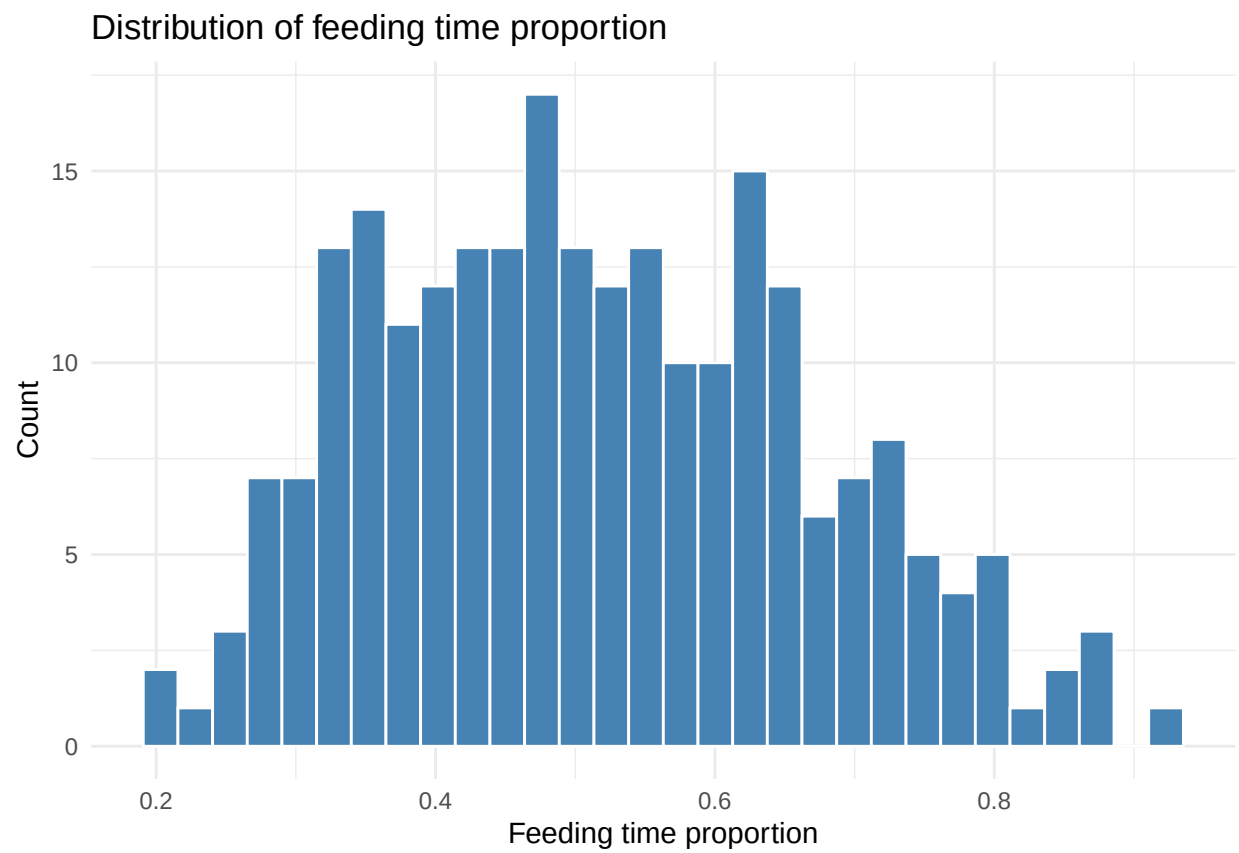
feeding_time_prop

Predictor variable:

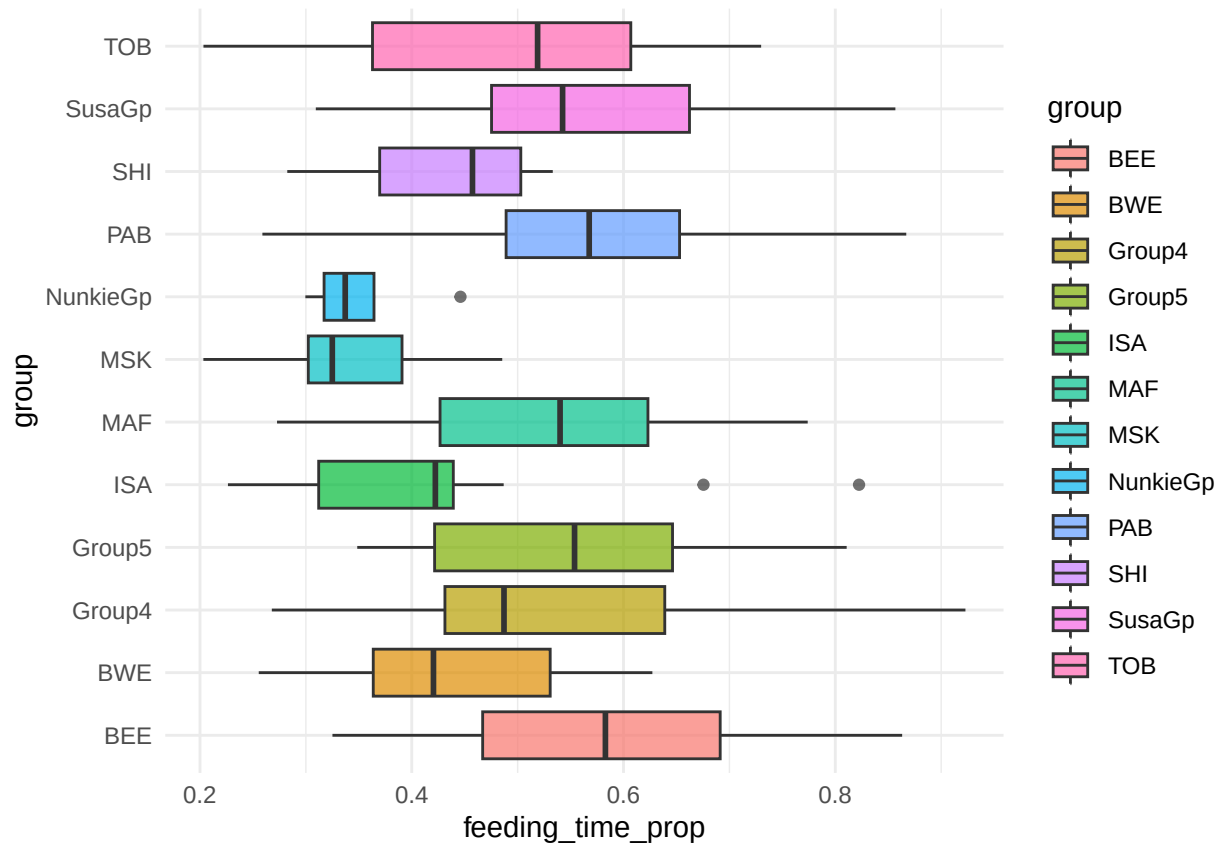
group

```
ggplot(data, aes(x = feeding_time_prop)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(
    title = "Distribution of feeding time proportion",
    x = "Feeding time proportion",
    y = "Count"
```

```
) +  
theme_minimal()
```



```
ggplot(data, aes(x = group, y = feeding_time_prop, fill = group)) +  
  geom_boxplot(alpha = 0.7) +  
  coord_flip() +  
  theme_minimal()
```



```
# =====
# Q: What is the response variable?
# Q: What type of variable is the response?
# Q: Why might proportions require special care in modeling?
# =====
```

Skills Application – Lab

8. Lab Challenge: Choose the Model Type

For this lab, your goal is **not** to run the final model yet.

Your goal is to look at a biological question, identify the response variable, diagnose the structure of the data, and decide what kind of model might be appropriate.

For each question below, answer:

1. What is the response variable?
2. What type of variable is the response?
3. What are the predictor variables?
4. Are the observations independent?
5. Is the dataset cross-sectional or longitudinal/repeated?
6. What plot would help you inspect the response?
7. What general model type might be appropriate?

8.1 Question 1

Do gorillas with more aggression events have higher cortisol?

Make at least one plot to inspect this question here

Your answer:

Response variable:

Response type:

Predictor variable(s):

Independent or repeated observations?

Cross-sectional or longitudinal?

Useful plot:

Possible model type:

8.2 Question 2

Are respiratory signs more likely in larger groups?

Make at least one plot to inspect this question here

Your answer:

Response variable:

Response type:

Predictor variable(s):

Independent or repeated observations?

Cross-sectional or longitudinal?

Useful plot:

Possible model type:

8.3 Question 3

Does feeding time proportion differ by group?

Make at least one plot to inspect this question here

Your answer:

Response variable:

Response type:

Predictor variable(s):

Independent or repeated observations?

Cross-sectional or longitudinal?

Useful plot:

Possible model type:

8.4 Optional: Your own biological question

Write one biological question using this dataset or your own dataset.

Your question:

```
# Make at least one plot to inspect your question here
```

Your answer:

Response variable:

Response type:

Predictor variable(s):

Independent or repeated observations?

Cross-sectional or longitudinal?

Useful plot:

Possible model type:

Summary

This week we:

- loaded and inspected a gorilla hormone + behavioral dataset
- identified continuous, count, binary, categorical, and proportion variables
- discussed cross-sectional vs. longitudinal data
- discussed independence vs. repeated measures
- diagnosed distributions using histograms, density plots, and QQ plots
- compared raw and log-transformed hormone values
- practiced linking biological questions to possible model types

Next week, we will start fitting and interpreting linear models more formally.

Optional - Additional Practice

```
# Count observations by group and sex
data %>%
  dplyr::count(group, sex, sort = TRUE)
```

```
## # A tibble: 19 x 3
##   group    sex      n
##   <chr>  <chr> <int>
## 1 PAB      M       40
## 2 Group4   F       24
## 3 SusaGp   F       24
## 4 BEE      F       16
## 5 Group5   F       16
## 6 PAB      F       16
## 7 BEE      M        8
## 8 BWE      F        8
## 9 Group5   M        8
## 10 ISA     F        8
## 11 ISA     M        8
## 12 MAF     F        8
```

```
## 13 MAF      M      8
## 14 MSK      M      8
## 15 NunkieGp M      8
## 16 SHI      F      8
## 17 SHI      M      8
## 18 TOB      F      8
## 19 TOB      M      8
```

```
# Summarize hormone values by sex
```

```
data %>%
  dplyr::group_by(sex) %>%
  dplyr::summarise(
    mean_cortisol = mean(cortisol_ng_ml, na.rm = TRUE),
    median_cortisol = median(cortisol_ng_ml, na.rm = TRUE),
    mean_testosterone = mean(testosterone_ng_ml, na.rm = TRUE),
    median_testosterone = median(testosterone_ng_ml, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )
```

```
## # A tibble: 2 x 6
##   sex   mean_cortisol median_cortisol mean_testosterone median_testosterone
##   <chr>         <dbl>         <dbl>         <dbl>         <dbl>
## 1 F             16.0             15.1             4.60           3.97
## 2 M             20.7             18.2            10.5           9.49
## # i 1 more variable: n <int>
```

```
# Summarize feeding time by group
```

```
data %>%
  dplyr::group_by(group) %>%
  dplyr::summarise(
    mean_feeding_time = mean(feeding_time_prop, na.rm = TRUE),
    median_feeding_time = median(feeding_time_prop, na.rm = TRUE),
    n = dplyr::n(),
    .groups = "drop"
  ) %>%
  dplyr::arrange(desc(mean_feeding_time))
```

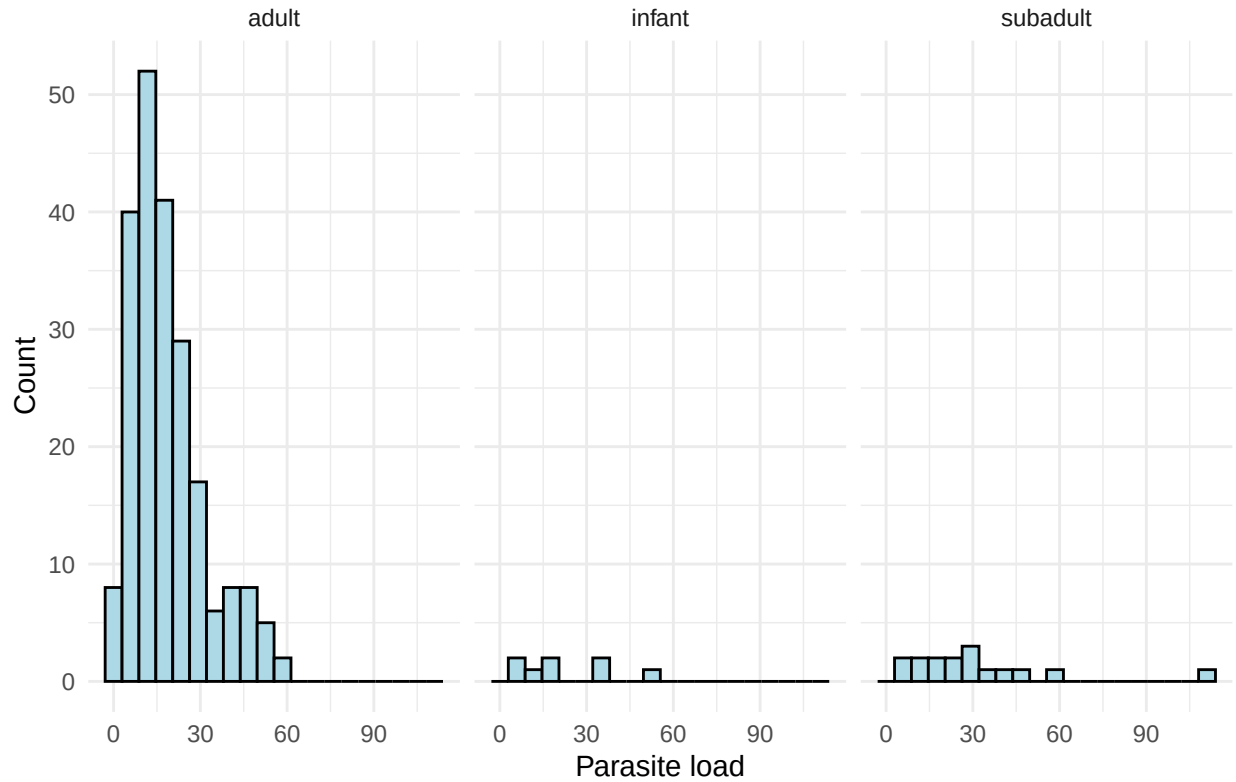
```
## # A tibble: 12 x 4
##   group   mean_feeding_time median_feeding_time     n
##   <chr>         <dbl>         <dbl> <int>
## 1 BEE             0.572             0.583    24
## 2 PAB             0.566             0.568    56
## 3 SusaGp         0.557             0.542    24
## 4 Group5         0.547             0.554    24
## 5 Group4         0.534             0.487    24
## 6 MAF             0.531             0.540    16
## 7 TOB             0.491             0.519    16
## 8 BWE             0.440             0.420     8
## 9 SHI             0.429             0.457    16
## 10 ISA            0.416             0.422    16
## 11 NunkieGp       0.347             0.337     8
## 12 MSK            0.342             0.325     8
```

```
# Density plot of log testosterone by sex
ggplot(data, aes(x = log_testosterone_ng_ml, fill = sex)) +
  geom_density(alpha = 0.5) +
  labs(
    title = "Log testosterone distribution by sex",
    x = "Log testosterone",
    y = "Density"
  ) +
  theme_minimal()
```



```
# Parasite load histogram by age class
ggplot(data, aes(x = parasite_load)) +
  geom_histogram(bins = 20, fill = "lightblue", color = "black") +
  facet_wrap(~ age_class) +
  labs(
    title = "Parasite load by age class",
    x = "Parasite load",
    y = "Count"
  ) +
  theme_minimal()
```

Parasite load by age class



```
# Identify individuals with repeated measurements
data %>%
  dplyr::count(gorilla_id, gorilla_name, sort = TRUE)
```

```
## # A tibble: 30 x 3
##   gorilla_id gorilla_name     n
##   <chr>      <chr>         <int>
## 1 G001      Gasore             8
## 2 G002      Poppy              8
## 3 G003      Ishimwe            8
## 4 G004      Umuyaga            8
## 5 G005      Ubusugire          8
## 6 G006      Mushya             8
## 7 G007      Chundura           8
## 8 G008      Twihangane         8
## 9 G009      Gitebe             8
## 10 G010      Petula             8
## # i 20 more rows
```